MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, NAND GATES

MONOLITHIC SILICON, POSITIVE LOGIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quadruple 2-Input NAND gate
02	Dual 4-Input NAND gate
03	Triple 3-Input NAND gate
51	Quadruple 2-Input NAND gate
52	Dual 4-Input NAND gate
53	Triple 3-Input NAND gate

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
X	F-1 (14-lead, 1/4" x 1/4"), flat package, except A dimension = 0.1" (2.54 mm) maximum
Y	F-2 (14-lead, 1/4" x 3/8"), flat package, except A dimension = 0.1" (2.54 mm) maximum

NOTES:
1. As an exception to 3.5.6.2.3 of MIL-M-38510, for case outlines X and Y only, the leads of bottom brazed ceramic packages (i.e., configuration 2 of case outlines F-1 or F-2) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e., finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.
2. For bottom or side brazed packages, case outlines X and Y only, the S_1 dimension may go to .000 inch (.00 mm) mimimum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: National Aeronautics and Space Administration, George C. Marshall Space Flight Center, ATTN: EG02 Marshall Space Flight Center, Alabama 35812, using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage (V_{DD}-V_{SS}):	
Device types 01, 02, and 03	-0.5 V to +15.5 V
Device types 51, 52, and 53	-0.5 V to +18 V
Input current (each input)	±10 mA
Input voltage range	(V_{SS}-0.5 V) ≤ V_I ≤ (V_{DD} + 0.5 V)
Storage temperature range	-65°C to +175°C
Maximum power dissipation (P_D)	200 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case	(See MIL-M-38510, appendix C)
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Device types 01, 02, and 03:	
Supply voltage (V_{DD}-V_{SS})	4.5 V dc to 12.5 V dc
Input low voltage range (V_{IL})	0-0.85 V dc @ V_{DD} = 5 V; 0-2.0 V dc @ V_{DD} = 10 V 0-2.1 V dc @ V_{DD} = 12.5 V
Input high voltage range (V_{IH})	3.95-5.0 V dc @ V_{DD} = 5 V; 8-10 V dc @ V_{DD} = 10 V; 10.0-12.5 V dc @ V_{DD} = 12.5 V
Device types 51, 52, and 53:	
Supply voltage (V_{DD}-V_{SS})	4.5 V dc to 15 V dc
Input low voltage range (V_{IL})	0-1.5 V dc @ V_{DD} = 5 V dc, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}, 0-2.0 V dc @ V_{DD} = 10 V dc. 0-4.0 V dc @ V_{DD} 15 V dc.
Input high voltage range (V_{IH})	3.5-5.0 V dc @ V_{DD} = 5 V dc, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}, 8-10 V dc @ V_{DD} = 10 V dc 11.0-15.0 V dc @ V_{DD} = 15 V dc
Ambient operating temperature (T_A)	-55°C to +125°C
Load capacitance	50 pF maximum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However, the resin used shall be Dupont 5504 Conductive Silver Paste, or equivalent which is cured at 200°C ±10°C for a minimum of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.6.1. Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.5 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 36 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute lines 1 through 7 of table II herein.

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b. Burn-in (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 3, or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 4 or equivalent. Ambient temperture (T_A) shall be 125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The class S devices PDA shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining PDA.

c. The class B devices PDA shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics after burn-in exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D and E inspections (see 4.4.1 through 4.4.5).

4.3.1 Qualification extension. When authorized by the qualifying activity, if a manufacturer qualifies to a 51, 52, or 53 device type which is manufactured identically to a 01, 02, or 03 device type on this specification, then the 01, 02, or 03 device type may be part I qualified by conducting only group A electrical tests and any electricals specified as additional group C subgroups and submitting data in accordance with MIL-M-38510, appendix D (i.e. groups A, B, C, D, and E tests are not required).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_I measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ V_{SS} = 0 V -55°C ≤ T_A ≤ +125°C unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Positive clamping input to V_{DD}	$V_{IC(pos)}$	T_A = 25°C, V_{DD} = GND, V_{SS} = Open, Output = Open, I_I = 1 mA		All	---	1.5	V
Negative clamping input to V_{SS}	$V_{IC(neg)}$	T_A = 25°C, V_{DD} = Open, V_{SS} = GND, Output = Open, I_I = -1 mA		All	---	-6	V
Quiescent supply current	I_{SS}	Any combinations of inputs	V_{DD} = 15 V	01,02,03	---	-750	nA
			V_{DD} = 18 V	51,52,53			
High level output voltage	V_{OH1}	V_{DD} = 5 V, I_{OH} = -0.175 mA Any one input = V_{IL} (see table III)		All	4.2	---	V
	V_{OH2}	V_{DD} = 5 V, I_{OH} = 0 Any one input = V_{IL} (see table III)		All	4.95	---	V
	V_{OH3}	V_{DD} = 12.5 V, I_{OH} = 0 Any one input = V_{IL} (see table III)		All	11.25	---	V
	V_{OH4}	V_{DD} = 15 V, I_{OH} = 0		51,52,53	14.95	---	V
Low level output voltage	V_{OL1}	V_{DD} = 5 V, I_{OL} = 0.175 mA I_{OL} = 0.085 mA All inputs = V_{IH} (see table III)		01 02,03	--- ---	0.5 0.7	V V
	V_{OL2}	V_{DD} = 5 V, I_{OL} = 0 All inputs = V_{IH} (see table III)		All	0.05	---	V
	V_{OL3}	V_{DD} = 12.5 V, I_{OL} = 0 All inputs = V_{IH} (see table III)		All	1.25	---	V
	V_{OL4}	V_{DD} = 15 V, I_{OL} = 0		51,52,53		0.05	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V -55°C ≤ T_A ≤ +125°C unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Input high voltage	V_{IH1}	V_{DD} = 5 V V_O = 0.5 V $\|I_O\|$ ≤1 µA		51,52, 53	3.5		V
	V_{IH2}	V_{DD} = 10 V V_O = 1.0 V $\|I_O\|$ ≤1 µA		51,52, 53	7.0		V
	V_{IH3}	V_{DD} = 15 V V_O = 1.5 V $\|I_O\|$ ≤1 µA		51,52, 53	11.0		V
Input low voltage	V_{IL1}	V_{DD} = 5 V V_O = 4.5 V $\|I_O\|$ ≤1 µA		51,52, 53		1.5	V
	V_{IL2}	V_{DD} = 10 V V_O = 9.0 V $\|I_O\|$ ≤1 µA		51,52, 53		3.0	V
	V_{IL3}	V_{DD} = 15 V V_O = 13.5 V $\|I_O\|$ ≤1 µA		51,52, 53		4.0	V
Output low (sink) current	I_{OL1}	V_{DD} = 5 V V_{IN} = 5 V V_{OL} = 0.4 V		51,52, 53	0.36		mA
	I_{OL2}	V_{DD} = 15 V V_{IN} = 15 V V_{OL} = 1.5 V		51,52, 53	2.4		mA
Output high (source) current	I_{OH1}	V_{DD} = 5 V Any one input = V_{SS} All other inputs = V_{DD} V_{OH} = 4.6 V		51,52, 53	-0.36		mA
	I_{OH2}	V_{DD} = 15 V Any one input = V_{SS} All other inputs = V_{DD} V_{OH} = 13.5 V		51,52, 53	-2.4		mA
Input leakage current 2/	I_{IH}	Measure inputs sequentially	V_{DD} = 15 V	01,02, 03		45	nA
			V_{DD} = 18 V	51,52, 53			
	I_{IL}	Measure inputs sequentially	V_{DD} = 15 V	01,02, 03		-45	nA
			V_{DD} = 18 V	51,52, 53			

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V -55°C ≤ T_A ≤ +125°C unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Input test voltage	V_{ZAP}	C_1 = 100 pF, R_Z = 1.5 kΩ (see 4.5.3)	All	400		V
Input capacitance	C_I	V_{DD} = 0 V, f = 1 MHz T_A = 25°C	All		12	pF
Propagation delay time high to low level	t_{PHL}	V_{DD} = 5 V, C_L = 50 pF (see figure 5)	01 02 03	15 15 15	300 490 265	ns
			51 52 53	10 16 9	300 490 265	
Propagation delay time low to high level	t_{PLH}		01,03 02	15 15	225 375	ns
			51,53 52	8 13	225 375	
Transition time high to low level	t_{THL}		01 02 03	25 40 25	450 825 375	ns
			51 52 53	15 28 13	450 825 375	
Transition time low to high level	t_{TLH}		01 02 03	30 35 30	450 640 450	ns
			51 52 53	15 21 12	450 640 450	

1/ Complete terminal conditions shall be as specified in table III.

2/ Input current at one input node.

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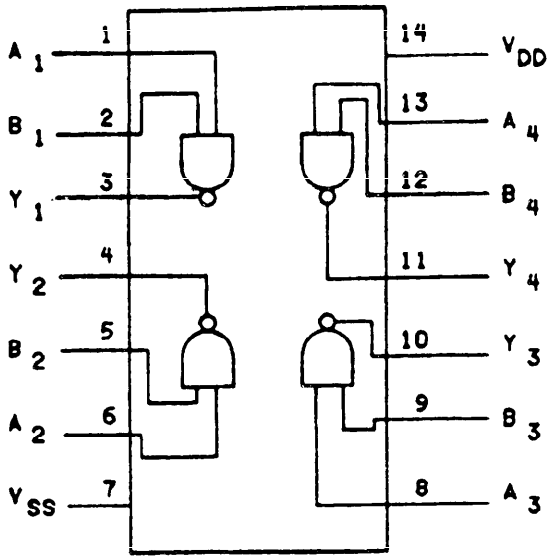
TABLE II. Burn-in and electrical test requirements.

Line no.	Applicable tests and MIL-STD-883 test method	Class S device 3/				Class B devices 3/			
		Ref. par.	Table 2/ III sub-groups	Table 1/ IV delta limits	Test circuit figure	Ref. par.	Table 2/ III sub-groups	Table 1/ IV delta limits	Test circuit figure
1	Interim electrical parameters (method 5004)		1				1		
2	Static burn-in I (method 1015)	4.2b 4.5.2			3				
3	Same as line 1		1	Δ					
4	Static burn-in II (method 1015)	4.2b 4.5.2			3	4.2b 4.5.2	4/		3
5	Same as line 1	4.2d	1*	Δ		4.2d	1*	Δ	
6	Dynamic burn-in (method 1015)	4.2b 4.5.2			4				
7	Same as line 1	4.2d	1*	Δ					
8	Final electrical parameters (method 5004)		1*,2, 3,9				1*,2, 3,9		
9	Group A end-point electrical parameters (method 5005)	4.4.1b	1,2,3,4, 9,10,11			4.4.1b	1,2,3, 4,9		
10	Group B end-point electical parameters (method 5005)	4.4.2	1,2,3, 9,10,11	Δ					
11	Group C end-point electrical parameters (method 5005)					4.4.3	1,2,3	Δ	
12	Additional group C tests, (method 5005)					4.4.3c	10, 11		
13	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3			4.4.4	1,2,3		

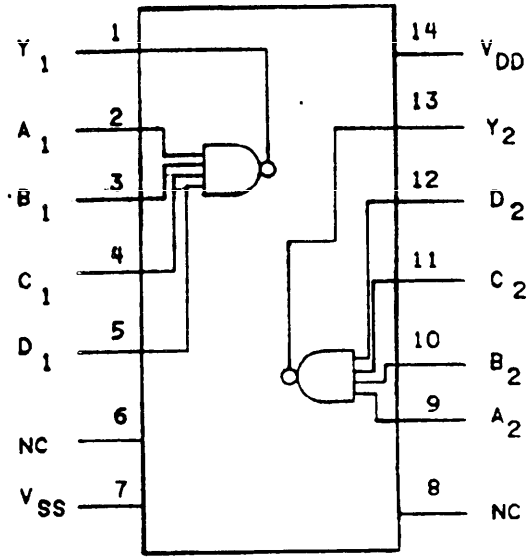
1/ (Δ) indicates delta limit shall be required on table III subgroup 1, where specified, and delta values shall be computed with reference to the previous interim electrical parameters.
2/ (*) indicates PDA applies to subgroup 1 (see 4.2.1).
3/ Blank spaces indicate tests are not applicable.
4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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Device type 01 and 51
Cases A, C, D, X and Y



Device types 02 and 52
Cases A, C, D, X and Y



Device types 03 and 53
Cases A, C, D, X and Y

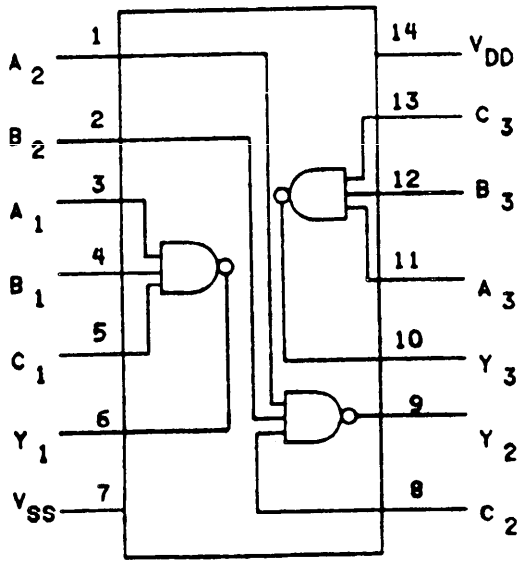


FIGURE 1. Logic diagrams and terminal connections.

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Device types 01 and 51

Truth table each gate		
Input		Output
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

Positive logic $Y = \overline{AB}$

Device types 02 and 52

Truth table each gate				
Input				Output
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

Positive logic $Y = \overline{ABCD}$

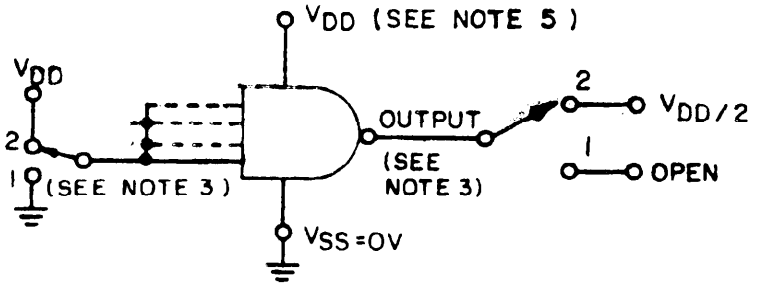
Device types 03 and 53

Truth table each gate			
Input			Output
A	B	C	Y
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	L

Positive logic $Y = \overline{ABC}$

FIGURE 2. Truth tables and logic equations.

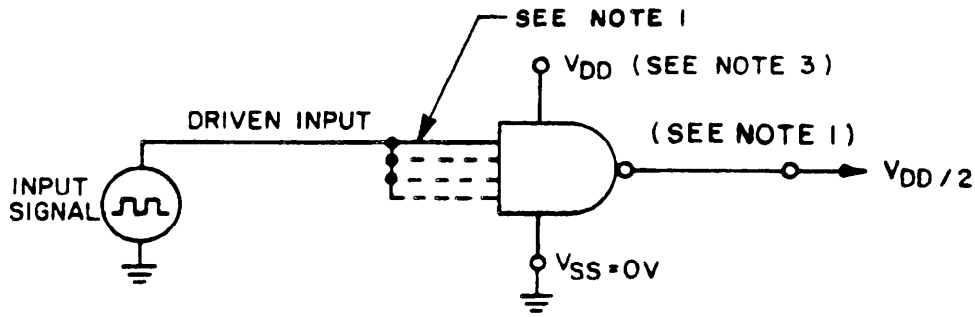
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NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ or 47 kΩ. The actual measured value of the resistor selected shall not exeed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.
5. V_{DD} = 12.5 V minimum, 15 V maximum for device 01, 02, 03.
   V_{DD} = 15 V minimum, 18 V maximum for device 51, 52, 53.
   $V_{DD/2} = V_{DD/2} \pm 1.0$ V.
   V_{SS} = 0 V.

FIGURE 3. Static burn-in test circuits.

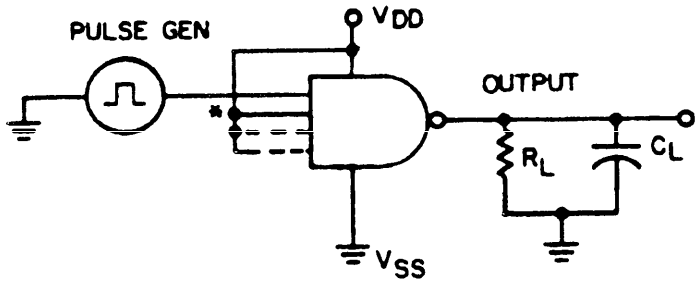
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NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ or 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz < PRR < 1 MHz.
   c. t_{TLH} and t_{THL} < 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5, -10% V_{DD}.
3. For device types 01 and 51:
   a. Connect pins 14, 2, and 11 together.
   b. Connect pins 7, 4, and 9 together.
   c. Connect pins 13 and 8 together.
   d. Connect pins 1 and 5 together.
4. V_{DD} = 12.5 V minimum, 15.0 V maximum for device 01, 02, 03, and 04.
   V_{DD} = 15.0 V minimum, 18.0 V maximum for device 51, 52, 53, and 54.
   $V_{DD/2}$ = $V_{DD/2}$ ±1.0 V.
   V_{SS} = 0.0 V.

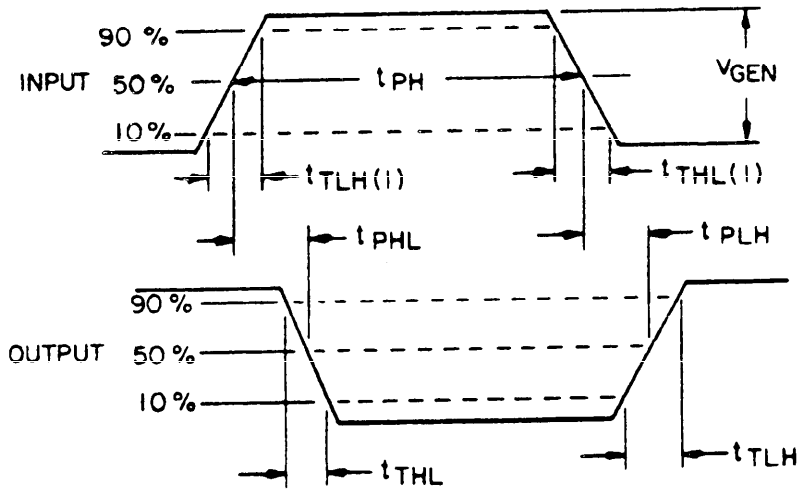
FIGURE 4. Dynamic burn-in and operating life test circuit.

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*NOTE: All unused inputs must be tied to V_{DD}

R_L = 200 kΩ ±10%
C_L = 50 pF ±5
(Includes wiring and probe capacitance)



DYNAMIC TEST WAVEFORMS

Input pulse

$V_{GEN} = V_{DD} \pm 1.0\%$
$t_{PH} = 1.0 \pm 0.1\ \mu s$
$t_{THL(1)} = t_{TLH(1)} = 10 \pm 2$ ns
PRR = 200 kHz

FIGURE 5. Switching time test circuit and waveforms.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1_A	1_B	1_Y	2_Y	2_B	2_A	V_{SS}	3_A	3_B	3_Y	4_Y	4_B	4_A	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$ 1/		1 2 3 4 5 6 7 8	1 mA	1 mA			1 mA	1 mA		1 mA	1 mA			1 mA	1 mA	GND " " " " " " "	1_A 1_B 2_B 2_A 3_A 3_B 4_B 4_A		1.5 " " " " " " "					V " " " " " " "
$V_{IC(neg)}$ 1/		9 10 11 12 13 14 15 16	-1 mA	-1 mA			-1 mA	-1 mA	GND " " " " " " "	-1 mA	-1 mA			-1 mA	-1 mA		1_A 1_B 2_B 2_A 3_A 3_B 4_B 4_A		-6.0 " " " " " " "					" " " " " " " "
I_{SS} 1/	3005 "	17 18 19	GND 15.0 V "	15.0 V " GND			15.0 V " GND	GND 15.0 V "	" " "	GND 15.0 V "	15.0 V " GND			15.0 V " GND	GND 15.0 V "	15.0 V " "	V_{SS} "		-25.0 " "		-750.0 " "			mA " "
V_{OH1}	3006 " " " " " " "	20 21 22 23 24 25 26 27	V_{IL1} 5.0 V GND " " " " "	5.0 V V_{IL1} GND " " " " "	I_{OH1}	I_{OH1}	" " V_{IL1} 5.0 V GND " " "	GND " 5.0 V V_{IL1} GND " " "	" " " " " " " "	GND " " " V_{IL1} 5.0 V GND "	" " " " 5.0 V V_{IL1} GND "	I_{OH1}	I_{OH1}	" " " " " V_{IL1} 5.0 V	GND " " " " 5.0 V V_{IL1}	5.0 V " " " " " " "	1_Y 1_Y 2_Y 2_Y 3_Y 3_Y 4_Y 4_Y	4.2 " " " " " " "		4.2 " " " " " " "		4.2 " " " " " " "		V " " " " " " "
V_{OH2}	" " " " " " "	28 29 30 31 32 33 34 35	V_{IL1} 5.0 V GND " " " " "	5.0 V V_{IL1} GND " " " " "			" " V_{IL1} 5.0 V GND " " "	" " 5.0 V V_{IL1} GND " " "	" " " " " " " "	" " " " V_{IL1} 5.0 V GND "	" " " " 5.0 V V_{IL1} GND "	" " " " " " " "	" " " " " V_{IL1} 5.0 V	GND " " " " 5.0 V V_{IL1}	" " " " " " " "	1_Y 1_Y 2_Y 2_Y 3_Y 3_Y 4_Y 4_Y	4.95 " " " " " " "		4.95 " " " " " " "		4.95 " " " " " " "		" " " " " " " "
V_{OH3}	" " " " " " "	36 37 38 39 40 41 42 43	V_{IL2} 12.5 V GND " " " " "	12.5 V V_{IL2} GND " " " " "			" " V_{IL2} 12.5 V GND " " "	" " 12.5 V V_{IL2} GND " " "	" " " " " " " "	" " " " V_{IL2} 12.5 V GND "	" " " " 12.5 V V_{IL2} GND "	" " " " " " " "	" " " " " V_{IL2} 12.5 V	GND " " " " 12.5 V V_{IL2}	12.5 V " " " " " " "	1_Y 1_Y 2_Y 2_Y 3_Y 3_Y 4_Y 4_Y	11.25 " " " " " " "		11.25 " " " " " " "		11.25 " " " " " " "		" " " " " " " "
V_{OL1}	3007 " " "	44 45 46 47	V_{IH1} GND " "	V_{IH1} GND " "	I_{OL1}	I_{OL1}	" V_{IH1} GND "	" V_{IH1} GND "	" " " "	" " V_{IH1} GND	" " V_{IH1} GND	I_{OL1}	I_{OL1}	GND " " V_{IH1}	GND " " V_{IH1}	5.0 V " " "	1_Y 2_Y 3_Y 4_Y		0.5 " " "		0.5 " " "		0.5 " " "	" " " "

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1_A	1_B	1_Y	2_Y	2_B	2_A	V_{SS}	3_A	3_B	3_Y	4_Y	4_B	4_A	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OL2}	3007	48 49 50 51	V_{IH1} GND "	V_{IH1} GND "			GND V_{IH1} GND	GND V_{IH1} GND	GND " "	GND " V_{IH1} GND	GND " V_{IH1} GND			GND " " V_{IH1}	GND " " V_{IH1}	5.0 V " " "	1Y 2Y 3Y 4Y		0.05 " " "		0.05 " " "		0.05 " " "	V " " "
V_{OL3}	" " " "	52 53 54 55	V_{IH2} GND "	V_{IH2} GND "			" V_{IH2} GND	" V_{IH2} GND	" " "	" " V_{IH2} GND	" " V_{IH2} GND			GND " " V_{IH2}	GND " " V_{IH2}	12.5 V " " "	1Y 2Y 3Y 4Y		1.25 " " "		1.25 " " "		1.25 " " "	" " " "
I_{IH1} 2/	3010	56	15.0 V	15.0 V			15.0 V	15.0 V	"	15.0 V	15.0 V			15.0 V	15.0 V	15.0 V	All together		8.0					mA
I_{IH2}	" " " " " " " "	57 58 59 60 61 62 63 64	" " " " " " " "	" " " " " " " "			" " " " " " " "	" " " " " " " "	" " " " " " " "	" " " " " " " "	" " " " " " " "			" " " " " " " "	" " " " " " " "	" " " " " " " "	1_A 1_B 2_B 2_A 3_A 3_B 4_B 4_A		1.0 " " " " " " "		45.0 " " " " " " "			" " " " " " " "
I_{IL1} 2/	3009	65	GND	GND			GND	GND	"	GND	GND			GND	GND	"	All together		-8.0					"
I_{IL2}	" " " " " " " "	66 67 68 69 70 71 72 73	" 15.0 V " " " " " "	15.0 V GND 15.0 V " " " " "			15.0 V " GND 15.0 V " " " "	15.0 V " GND 15.0 V " " " "	" " " " " " " "	15.0 V " " GND 15.0 V " " "	15.0 V " " GND 15.0 V " " "			15.0 V " " " " " GND 15.0 V	15.0 V " " " " " GND "	" " " " " " " "	1_A 1_B 2_B 2_A 3_A 3_B 4_B 4_A		-1.0 " " " " " " "		-45.0 " " " " " " "			" " " " " " " "
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_I	3012 " " " " " " "	74 75 76 77 78 79 80 81	F	F			F	F	GND " " " " " " "	F	F			F	F	GND " " " " " " "	1_A 1_B 2_B 2_A 3_A 3_B 4_B 4_A		12.0 " " " " " " "					pF " " " " " " "
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5 " " " " " "	82 83 84 85 86 87 88 89	IN 5.0 V " " " " " "	5.0 V IN 5.0 V " " " " "	OUT OUT	OUT OUT	5.0 V " IN 5.0 V " " " "	5.0 V " IN 5.0 V " " " "	GND " " " " " " "	5.0 V " " IN 5.0 V " " "	5.0 V " " IN 5.0 V 5.0 V " "	OUT OUT	OUT OUT	5.0 V " " " IN 5.0 V " "	5.0 V " " " IN " " "	5.0 V " " " " " " "	1_A to 1_Y 1_B to 1_Y 2_B to 2_Y 2_A to 2_Y 3_A to 3_Y 3_B to 3_Y 4_B to 4_Y 4_A to 4_Y		20 " " " " " " "	200 " " " " " " "	30 " " " " " " "	300 " " " " " " "	15 " " " " " " "	200 " " " " " " "	ns " " " " " " "

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	1_A	1_B	1_Y	2_Y	2_B	2_A	V_{SS}	3_A	3_B	3_Y	4_Y	4_B	4_A	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PLH}	3003 Fig. 5	90	IN	5.0 V	OUT		5.0 V	5.0 V	GND	5.0 V	5.0 V			5.0 V	5.0 V	5.0 V	1_A to 1_Y	20	150	30	225	15	150	ns
		91	5.0 V	IN	OUT												1_B to 1_Y							
		92		5.0 V		OUT	IN										2_B to 2_Y							
		93				OUT	5.0 V	IN									2_A to 2_Y							
		94						5.0 V		IN		OUT					3_A to 3_Y							
		95								5.0 V	IN	OUT					3_B to 3_Y							
		96									5.0 V		OUT	IN			4_B to 4_Y							
		97											OUT	5.0 V	IN		4_A to 4_Y							
t_{THL}	3004 Fig. 5	98	IN		OUT										5.0 V		1_Y	30	300	45	450	25	300	ns
		99	5.0 V	IN	OUT												1_Y							
		100		5.0 V		OUT	IN										2_Y							
		101				OUT	5.0 V	IN									2_Y							
		102						5.0 V		IN		OUT					3_Y							
		103								5.0 V	IN	OUT					3_Y							
		104									5.0 V		OUT	IN			4_Y							
		105											OUT	5.0 V	IN		4_Y							
t_{TLH}		106	IN		OUT										5.0 V		1_Y	40		60		30		
		107	5.0 V	IN	OUT												1_Y							
		108		5.0 V		OUT	IN										2_Y							
		109				OUT	5.0 V	IN									2_Y							
		110						5.0 V		IN		OUT					3_Y							
		111								5.0 V	IN	OUT					3_Y							
		112									5.0 V		OUT	IN			4_Y							
		113											OUT	5.0 V	IN		4_Y							

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1_Y	1_A	1_B	1_C	1_D	NC	V_{SS}	NC	2_A	2_B	2_C	2_D	2_Y	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$ 1/		1 2 3 4 5 6 7 8		1 mA	1 mA	1 mA	1 mA				1 mA	1 mA	1 mA	1 mA		GND " " " " " "	1_A 1_B 1_C 1_D 2_A 2_B 2_C 2_D		1.5 " " " " " " "					V " " " " " " "
$V_{IC(neg)}$ 1/		9 10 11 12 13 14 15 16		-1 mA	-1 mA	-1 mA	-1 mA		GND " " " " " "		-1 mA	-1 mA	-1 mA	-1 mA			1_A 1_B 1_C 1_D 2_A 2_B 2_C 2_D		-6.0 " " " " " " "					" " " " " " " "
I_{SS} 1/	3005 " " "	17 18 19 20 21		GND 15.0 V " " "	15.0 V GND 15.0 V " "	15.0 V " GND 15.0 V "	15.0 V " " GND 15.0 V		" " " " " "		GND 15.0 V " " "	15.0 V GND 15.0 V " "	15.0 V " GND 15.0 V "	15.0 V " " GND 15.0 V		15.0 V " " " "	V_{SS} " " " "		-25.0 " " " "		-750.0 " " "			nA " " "
V_{ON1}	3006 " " " " " " "	22 23 24 25 26 27 28 29	I_{OH1} " "	V_{IL1} 5.0 V " " GND " " "	5.0 V V_{IL1} 5.0 V " GND " " "	5.0 V " V_{IL1} 5.0 V GND " " "	5.0 V " " V_{IL1} GND " " "		" " " " " " " "		GND " " V_{IL1} 5.0 V " " "	GND " " 5.0 V V_{IL1} 5.0 V " "	GND " " 5.0 V " V_{IL1} 5.0 V "	GND " " 5.0 V " " V_{IL1} "	I_{OH1} " "	5.0 V " " " " " " "	1_Y 1_Y 1_Y 1_Y 2_Y 2_Y 2_Y 2_Y	4.2 " " " " " " "		4.2 " " " " " " "		4.2 " " " " " " "		V " " " " " " "
V_{OH2}	" " " " " " "	30 31 32 33 34 35 36 37		V_{IL1} 5.0 V " " GND " " "	5.0 V V_{IL1} 5.0 V " GND " " "	5.0 V " V_{IL1} 5.0 V GND " " "	5.0 V " " V_{IL1} GND " " "		" " " " " " " "		GND " " V_{IL1} 5.0 V " " "	GND " " 5.0 V V_{IL1} 5.0 V " "	GND " " 5.0 V " V_{IL1} 5.0 V "	GND " " 5.0 V " " V_{IL1} "		" " " " " " " "	1_Y 1_Y 1_Y 1_Y 2_Y 2_Y 2_Y 2_Y	4.95 " " " " " " "		4.95 " " " " " " "		4.95 " " " " " " "		" " " " " " " "
V_{ON3}	" " " " " " "	38 39 40 41 42 43 44 45		V_{IL2} 12.5 V " " GND " " "	12.5 V V_{IL2} 12.5 V " GND " " "	12.5 V " V_{IL2} 12.5 V GND " " "	12.5 V " " V_{IL2} GND " " "		" " " " " " " "		GND " " V_{IL2} 12.5 V " " "	GND " " 12.5 V V_{IL2} 12.5 V " "	GND " " 12.5 V " V_{IL2} 12.5 V "	GND " " 12.5 V " " V_{IL2} "		12.5 V " " " " " " "	1_Y 1_Y 1_Y 1_Y 2_Y 2_Y 2_Y 2_Y	11.25 " " " " " " "		11.25 " " " " " " "		11.25 " " " " " " "		" " " " " " " "
V_{OL1}	3007 "	46 47	I_{OL2}	V_{IH1} GND	V_{IH1} GND	V_{IH1} GND	V_{IH1} GND		" "		GND V_{IH1}	GND V_{IH1}	GND V_{IH1}	GND V_{IH1}	I_{OL2}	5.0 V "	1_Y 2_Y		0.7 "		0.7 "		0.7 "	"

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TABLE III. Group A inspection for device type 02 -Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1_Y	1_A	1_B	1_C	1_D	NC	V_{SS}	NC	2_A	2_B	2_C	2_D	2_Y	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OL2}	3007	48		V_{IH1} GND	V_{IH1} GND	V_{IH1} GND	V_{IH1} GND		GND		GND V_{IH1}	GND V_{IH1}	GND V_{IH1}	GND V_{IH1}		5.0 V	1_Y 2_Y		0.05		0.05		0.05	V
V_{OL3}	"	50		V_{IH2} GND	V_{IH2} GND	V_{IH2} GND	V_{IH2} GND		"		GND V_{IH2}	GND V_{IH2}	GND V_{IH2}	GND V_{IH2}		12.5 V	1_Y 2_Y		1.25		1.25		1.25	"
I_{IH1} 2/	3010	52		15.0 V	15.0 V	15.0 V	15.0 V		"		15.0 V	15.0 V	15.0 V	15.0 V		15.0 V	All together		8.0					nA
I_{IH2}	"	53		"	"	"	"		"		"	"	"	"		"	1_A		1.0		45.0			"
	"	54		"	"	"	"		"		"	"	"	"		"	1_B		"		"			"
	"	55		"	"	"	"		"		"	"	"	"		"	1_C		"		"			"
	"	56		"	"	"	"		"		"	"	"	"		"	1_D		"		"			"
	"	57		"	"	"	"		"		"	"	"	"		"	2_A		"		"			"
	"	58		"	"	"	"		"		"	"	"	"		"	2_B		"		"			"
	"	59		"	"	"	"		"		"	"	"	"		"	2_C		"		"			"
	"	60		"	"	"	"		"		"	"	"	"		"	2_D		"		"			"
I_{IL1} 2/	3009	61		GND	GND	GND	GND		"		GND	GND	GND	GND		"	All together		-8.0					"
I_{IL2}	"	62		"	15.0 V	15.0 V	15.0 V		"		15.0 V	15.0 V	15.0 V	15.0 V		"	1_A		-1.0		-45.0			"
	"	63		15.0 V	GND	"	"		"		"	"	"	"		"	1_B		"		"			"
	"	64		"	15.0 V	GND	"		"		"	"	"	"		"	1_C		"		"			"
	"	65		"	"	15.0 V	GND		"		"	"	"	"		"	1_D		"		"			"
	"	66		"	"	"	15.0 V		"		GND	"	"	"		"	2_A		"		"			"
	"	67		"	"	"	"		"		15.0 V	GND	"	"		"	2_B		"		"			"
	"	68		"	"	"	"		"		"	15.0 V	GND	"		"	2_C		"		"			"
	"	69		"	"	"	"		"		"	"	15.0 V	GND		"	2_D		"		"			"
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_f	3012	70		F					GND							GND	1_A		12.0					pF
	"	71			F				"							"	1_B		"					"
	"	72				F			"							"	1_C		"					"
	"	73					F		"							"	1_D		"					"
	"	74							"		F					"	2_A		"					"
	"	75							"			F				"	2_B		"					"
	"	76							"				F			"	2_C		"					"
	"	77							"					F		"	2_D		"					"
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5	78	OUT	IN	5.0 V	5.0 V	5.0 V		GND		5.0 V	5.0 V	5.0 V	5.0 V		5.0 V	1_Y	20	325	30	490	15	325	ns
	"	79	"	5.0 V	IN	"	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
	"	80	"	"	5.0 V	IN	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
	"	81	"	"	"	5.0 V	IN		"		"	"	"	"		"	2_Y	"	"	"	"	"	"	"
	"	82	"	"	"	"	5.0 V		"		IN	"	"	"	OUT	"	2_Y	"	"	"	"	"	"	"
	"	83	"	"	"	"	"		"		5.0 V	IN	"	"	"	"	2_Y	"	"	"	"	"	"	"
	"	84	"	"	"	"	"		"		"	5.0 V	IN	"	"	"	2_Y	"	"	"	"	"	"	"
	"	85	"	"	"	"	"		"		"	"	5.0 V	IN	"	"	2_Y	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02 -Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	1_Y	1_A	1_B	1_C	1_D	NC	V_{SS}	NC	2_A	2_B	2_C	2_D	2_Y	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PLH}	3003 Fig. 5	86	OUT	IN	5.0 V	5.0 V	5.0 V		GND		5.0 V	5.0 V	5.0 V	5.0 V		5.0 V	1_Y	20	250	30	375	15	250	ns
		87	"	5.0 V	IN	"	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	
		88	"	"	5.0 V	IN	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	
		89	"	"	"	5.0 V	IN		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	
		90	"	"	"	"	5.0 V		"		IN	"	"	"	OUT	"	2_Y	"	"	"	"	"	"	
		91	"	"	"	"	"		"		5.0 V	IN	IN	"	"	"	2_Y	"	"	"	"	"	"	
		92	"	"	"	"	"		"		"	5.0 V	5.0 V	IN	"	"	2_Y	"	"	"	"	"	"	
		93	"	"	"	"	"		"		"	"	"	"	"	"	2_Y	"	"	"	"	"	"	
t_{THL}	3004 Fig. 5	94	OUT	IN	"	"	"		"		"	"	"	5.0 V		"	1_Y	50	550	70	825	40	550	"
		95	"	5.0 V	IN	"	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	
		96	"	"	5.0 V	IN	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	
		97	"	"	"	5.0 V	IN		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	
		98	"	"	"	"	5.0 V		"		IN	"	"	"	OUT	"	2_Y	"	"	"	"	"	"	
		99	"	"	"	"	"		"		5.0 V	IN	IN	"	"	"	2_Y	"	"	"	"	"	"	
		100	"	"	"	"	"		"		"	5.0 V	5.0 V	IN	"	"	2_Y	"	"	"	"	"	"	
		101	"	"	"	"	"		"		"	"	"	"	"	"	2_Y	"	"	"	"	"	"	
t_{TLH}	"	102	OUT	IN	"	"	"		"		"	"	"	5.0 V		"	1_Y	40	425	60	640	35	425	"
		103	"	5.0 V	IN	"	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	
		104	"	"	5.0 V	IN	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	
		105	"	"	"	5.0 V	IN		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	
		106	"	"	"	"	5.0 V		"		IN	"	"	"	OUT	"	2_Y	"	"	"	"	"	"	
		107	"	"	"	"	"		"		5.0 V	IN	IN	"	"	"	2_Y	"	"	"	"	"	"	
		108	"	"	"	"	"		"		"	5.0 V	5.0 V	IN	"	"	2_Y	"	"	"	"	"	"	
		109	"	"	"	"	"		"		"	"	"	"	"	"	2_Y	"	"	"	"	"	"	

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	2_A	2_B	1_A	1_B	1_C	1_Y	V_{SS}	2_C	2_Y	3_Y	3_A	3_B	3_C	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$ 1/		1	1 mA													GND	2_A		1.5					V
		2		1 mA												"	2_B		"					"
		3			1 mA					1 mA						"	2_C		"					"
		4				1 mA										"	1_A		"					"
		5					1 mA									"	1_B		"					"
		6														"	1_C		"					"
		7											1 mA			"	3_A		"					"
		8												1 mA		"	3_B		"					"
		9													1 mA	"	3_C		"					"
$V_{IC(neg)}$ 1/		10	-1 mA						GND								2_A		-6.0					V
		11		-1 mA					"								2_B		"					"
		12			-1 mA				"	-1 mA							2_C		"					"
		13				-1 mA			"								1_A		"					"
		14					-1 mA		"								1_B		"					"
		15							"								1_C		"					"
		16							"				-1 mA				3_A		"					"
		17							"					-1 mA			3_B		"					"
		18							"						-1 mA		3_C		"					"
I_{SS} 1/	3005	19	GND	15.0 V	GND	15.0 V	15.0 V		"	15.0 V			GND	15.0 V	15.0 V	15.0 V	V_{SS}		-25.0		-750			mA
	"	20	15.0 V	"	15.0 V	"	"		"	"			15.0 V	"	"	"	"		"		"			"
	"	21	"	GND	"	GND	GND		"	GND			"	GND	GND	"	"		"		"			"
	"	22	"	15.0 V	"	15.0 V	"		"	"			"	15.0 V	"	"	"		"		"			"
V_{OH1}	3006	23	V_{IL1}	5.0 V	GND	GND	"		"	5.0 V	I_{OH1}		GND	GND	GND	5.0 V	2_Y	4.2		4.2		4.2		V
	"	24	5.0 V	V_{IL1}	"	"	"		"	"	"		"	"	"	"	2_Y	"		"		"		"
	"	25	"	5.0 V		"			"	V_{IL1}			"	"	"	"	2_Y	"		"		"		"
	"	26	GND	GND	V_{IL1}	5.0 V	5.0 V	I_{OH1}	"	GND			"	"	"	"	1_Y	"		"		"		"
	"	27	"	"	5.0 V	"	"	"	"	"			"	"	"	"	1_Y	"		"		"		"
	"	28	"	"	"	V_{IL1}	V_{IL1}	"	"	"			"	"	"	"	1_Y	"		"		"		"
	"	29	"	"	GND	5.0 V	GND	"	"	"			V_{IL1}	5.0 V	5.0 V	"	3_Y	"		"		"		"
	"	30	"	"	"	GND	"	"	"	"		I_{OH1}	5.0 V	V_{IL1}	V_{IL1}	"	3_Y	"		"		"		"
	"	31	"	"	"	"	"	"	"	"		"	"	5.0 V	"	"	3_Y	"		"		"		"
V_{OH2}	"	32	V_{IL1}	5.0 V	"	"	"		"	5.0 V			GND	GND	GND	5.0 V	2_Y	4.95		4.95		4.95		"
	"	33	5.0 V	V_{IL1}	"	"	"		"	"			"	"	"	"	2_Y	"		"		"		"
	"	34	"	5.0 V		"			"	V_{IL1}			"	"	"	"	2_Y	"		"		"		"
	"	35	GND	GND	V_{IL1}	5.0 V	5.0 V		"	GND			"	"	"	"	1_Y	"		"		"		"
	"	36	"	"	5.0 V	V_{IL1}	"		"	"			"	"	"	"	1_Y	"		"		"		"
	"	37	"	"	"	5.0 V	V_{IL1}		"	"			"	"	"	"	1_Y	"		"		"		"
	"	38	"	"	GND	GND	GND		"	"			V_{IL1}	5.0 V	5.0 V	"	3_Y	"		"		"		"
	"	39	"	"	"	"	"		"	"			5.0 V	V_{IL1}	V_{IL1}	"	3_Y	"		"		"		"
	"	40	"	"	"	"	"		"	"			"	5.0 V	"	"	3_Y	"		"		"		"
V_{OH3}	"	41	V_{IL2}	12.5 V	"	"	"		"	12.5 V			GND	GND	GND	12.5 V	2_Y	11.25		11.25		11.25		"
	"	42	12.5 V	V_{IL2}	"	"	"		"	"			"	"	"	"	2_Y	"		"		"		"
	"	43	"	12.5 V		"			"	V_{IL2}			"	"	"	"	2_Y	"		"		"		"
	"	44	GND	GND	V_{IL2}	12.5 V	12.5 V		"	GND			"	"	"	"	1_Y	"		"		"		"
	"	45	"	"	12.5 V	V_{IL2}	"		"	"			"	"	"	"	1_Y	"		"		"		"
	"	46	"	"	"	12.5 V	V_{IL2}		"	"			"	"	"	"	1_Y	"		"		"		"
	"	47	"	"	GND	GND	GND		"	"			V_{IL2}	12.5 V	12.5 V	"	3_Y	"		"		"		"
	"	48	"	"	"	"	"		"	"			12.5 V	V_{IL2}	"	"	3_Y	"		"		"		"
	"	49	"	"	"	"	"		"	"			"	12.5 V	V_{IL2}	"	3_Y	"		"		"		"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	2_A	2_B	1_A	1_B	1_C	1_Y	V_{SS}	2_C	2_Y	3_Y	3_A	3_B	3_C	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OL1}	3007	50 51 52	V_{IH1} GND	V_{IH1} GND	GND V_{IH1} GND	GND V_{IH1} GND	GND V_{IH1} GND	I_{OL1}	GND	V_{IH1} GND	I_{OL1}	I_{OL1}	GND V_{IH1}	GND V_{IH1}	GND V_{IH1}	5.0 V	2_Y 1_Y 3_Y		0.7		0.7		0.7	V
V_{OL2}		53 54 55	V_{IH1} GND	V_{IH1} GND	V_{IH1} GND	V_{IH1} GND	V_{IH1} GND			V_{IH1} GND			GND V_{IH1}	GND V_{IH1}	GND V_{IH1}		2_Y 1_Y 3_Y		0.05		0.05		0.05	
V_{OL3}		56 57 58	V_{IH2} GND	V_{IH2} GND	V_{IH2} GND	V_{IH2} GND	V_{IH2} GND			V_{IH2} GND			GND V_{IH2}	GND V_{IH2}	GND V_{IH2}	12.5 V	2_Y 1_Y 3_Y		1.25		1.25		1.25	
I_{IH1} 2/	3010	59	15.0 V	15.0 V	15.0 V	15.0 V	15.0 V			15.0 V			15.0 V	15.0 V	15.0 V	15.0 V	All together		9.0					nA
I_{IH2}		60 61 62 63 64 65 66 67 68															2_A 2_B 2_C 1_A 1_B 1_C 3_A 3_B 3_C		1.0		45.0			
I_{IL1} 2/		69	GND	GND	GND	GND	GND			GND			GND	GND	GND		All together		-9.0					
I_{IL2}		70 71 72 73 74 75 76 77 78	15.0 V	15.0 V GND 15.0 V	15.0 V GND 15.0 V	15.0 V GND 15.0 V	15.0 V GND 15.0 V			15.0 V GND 15.0 V			15.0 V GND 15.0 V	15.0 V GND 15.0 V	15.0 V GND		2_A 2_B 2_C 1_A 1_B 1_C 3_A 3_B 3_C		-1.0		-45.0			
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_I	3012	79 80 81 82 83 84 85 86 87	F	F	F	F	F		GND	F			F	F	F	GND	2_A 2_B 2_C 1_A 1_B 1_C 3_A 3_B 3_C		12.0					pF

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	2_A	2_B	1_A	1_B	1_C	1_Y	V_{SS}	2_C	2_Y	3_Y	3_A	3_B	3_C	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5	88	IN	5.0 V	5.0 V	5.0 V	5.0 V		GND	5.0 V	OUT		5.0 V	5.0 V	5.0 V	5.0 V	2_Y	20	175	30	265	15	175	ns
		89	5.0 V	IN	"	"	"		"	"	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		90	"	5.0 V	"	"	"		"	IN	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		91	"	"	IN	"	"	OUT	"	5.0 V	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		92	"	"	5.0 V	IN	"	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		93	"	"	"	5.0 V	IN	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		94	"	"	"	"	5.0 V	"	"	"	"	OUT	IN	"	"	"	3_Y	"	"	"	"	"	"	"
		95	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	"	"	3_Y	"	"	"	"	"	"	"
		96	"	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	"	3_Y	"	"	"	"	"	"	"
t_{PLH}	"	97	IN	"	"	"	"		"	"	OUT		"	"	5.0 V	"	2_Y	"	150	"	225	"	150	"
		98	5.0 V	IN	"	"	"		"	"	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		99	"	5.0 V	"	"	"		"	IN	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		100	"	"	IN	"	"	OUT	"	5.0 V	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		101	"	"	5.0 V	IN	"	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		102	"	"	"	5.0 V	IN	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		103	"	"	"	"	5.0 V	"	"	"	"	OUT	IN	"	"	"	3_Y	"	"	"	"	"	"	"
		104	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	"	"	3_Y	"	"	"	"	"	"	"
		105	"	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	"	3_Y	"	"	"	"	"	"	"
t_{THL}	3004 Fig. 5	106	IN	"	"	"	"		"	"	OUT		"	"	5.0 V	"	2_Y	30	250	45	375	25	250	"
		107	5.0 V	IN	"	"	"		"	"	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		108	"	5.0 V	"	"	"		"	IN	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		109	"	"	IN	"	"	OUT	"	5.0 V	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		110	"	"	5.0 V	IN	"	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		111	"	"	"	5.0 V	IN	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		112	"	"	"	"	5.0 V	"	"	"	"	OUT	IN	"	"	"	3_Y	"	"	"	"	"	"	"
		113	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	"	"	3_Y	"	"	"	"	"	"	"
		114	"	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	"	3_Y	"	"	"	"	"	"	"
t_{TLH}	"	115	IN	"	"	"	"		"	"	OUT		"	"	5.0 V	"	2_Y	40	300	60	450	30	300	"
		116	5.0 V	IN	"	"	"		"	"	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		117	"	5.0 V	"	"	"		"	IN	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		118	"	"	IN	"	"	OUT	"	5.0 V	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		119	"	"	5.0 V	IN	"	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		120	"	"	"	5.0 V	IN	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
		121	"	"	"	"	5.0 V	"	"	"	"	OUT	IN	"	"	"	3_Y	"	"	"	"	"	"	"
		122	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	"	"	3_Y	"	"	"	"	"	"	"
		123	"	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	"	3_Y	"	"	"	"	"	"	"

1/ Pins not designated may be "High" level logic, "Low" level logic, or open. Exceptions are as follows:

a. $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open.
b. $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open.
c. I_{SS} tests, the output terminal shall be open.

2/ The device manufacturer may at his option measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

3/

Symbol	V_{IH1}	V_{IL1}	V_{IH2}	V_{IL2}	I_{OH1}	I_{OL1}	I_{OL2}	F
Temperature								
T_A = 25°C	3.95 V	0.9 V	10.25 V	2.25 V	-0.25 mA	0.25 mA	0.12 mA	See 4.4.1.c
T_A = 125°C	3.85 V	0.65 V	10.00 V	1.95 V	-0.175 mA	0.175 mA	0.085 mA	
T_A = -55°C	4.05 V	0.95 V	10.50 V	2.40 V	-0.31 mA	0.31 mA	0.15 mA	

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TABLE III. Group A inspection for device type 51.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1_A	1_B	1_Y	2_Y	2_B	2_A	V_{SS}	3_A	3_B	3_Y	4_Y	4_B	4_A	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$ 1/		1 2 3 4 5 6 7 8	1 mA	1 mA			1 mA	1 mA		1 mA	1 mA			1 mA	1 mA	GND " " " " " " "	1_A 1_B 2_B 2_A 3_A 3_B 4_B 4_A		1.5 " " " " " " "					V " " " " " " "
$V_{IC(neg)}$ 1/		9 10 11 12 13 14 15 16	-1 mA	-1 mA			-1 mA	-1 mA	GND " " " " " " "	-1 mA	-1 mA			-1 mA	-1 mA		1_A 1_B 2_B 2_A 3_A 3_B 4_B 4_A		-6.0 " " " " " " "					" " " " " " " "
I_{SS} 1/	3005 "	17 18 19	GND 18.0 V "	18.0 V " GND			18.0 V " GND	GND 18.0 V "	" " "	GND 18.0 V "	18.0 V " GND			18.0 V " GND	GND 18.0 V "	18.0 V "	V_{SS} "		-25 "		-750 "			mA "
V_{OH4}	3006 " " " " " "	20 21 22 23 24 25 26 27	GND 15 V GND " " " " "	15 V GND " " " " " "			" " 15 V GND " " " "	" " 15 V GND " " " "	" " " " " " " "	" " " " " 15 V GND "	" " " " " 15 V GND "			" " " " " GND 15 V	GND " " " " 15 V GND	15 V " " " " " "	1_Y 1_Y 2_Y 2_Y 3_Y 3_Y 4_Y 4_Y	14.95 " " " " " " "		14.95 " " " " " " "		14.95 " " " " " " "		V " " " " " " "
V_{OL4}	3007 " "	28 29 30 31	15 V GND " "	15 V GND " "			" 15 V GND "	" 15 V GND "	" " " "	" " 15 V GND	" " 15 V GND			GND " " 15 V	" " " 15 V	" " "	1_Y 2_Y 3_Y 4_Y		0.05 " " "		0.05 " " "		0.05 " " "	" " " "
V_{IH1}		32 33 34 35	3.5 V GND " "	3.5 V GND " "			" 3.5 V GND "	" 3.5 V GND "	" " " "	" " 3.5 V GND	" " 3.5 V GND			GND " " 3.5 V	GND " " 3.5 V	5 V " "	1_Y 2_Y 3_Y 4_Y		0.5 " " "		0.5 " " "		0.5 " " "	" " " "
V_{IH2}		36 37 38 39	7.0 V GND " "	7.0 V GND " "			" 7.0 V GND "	" 7.0 V GND "	" " " "	" " 7.0 V GND	" " 7.0 V GND			GND " " 7.0 V	GND " " 7.0 V	10 V " "	1_Y 2_Y 3_Y 4_Y		1.0 " " "		1.0 " " "		1.0 " " "	" " " "
V_{IH3}		40 41 42 43	11 V GND " "	11 V GND " "			" 11 V GND "	" 11 V GND "	" " " "	" " 11 V GND	" " 11 V GND			GND " " 11 V	GND " " 11 V	15 V " "	1_Y 2_Y 3_Y 4_Y		1.5 " " "		1.5 " " "		1.5 " " "	" " " "

See footnotes at end of device type 53.

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1_A	1_B	1_Y	2_Y	2_B	2_A	V_{SS}	3_A	3_B	3_Y	4_Y	4_B	4_A	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IL1}		44	1.5 V	1.5 V			GND	GND	GND	GND	GND			GND	GND	5 V	1_Y	4.5		4.5		4.5		V
		45	3.5 V	"			"	"	"	"	"			"	"	"	1_Y	"		"		"		"
		46	1.5 V	3.5 V			"	"	"	"	"			"	"	"	1_Y	"		"		"		"
		47	GND	GND			1.5 V	1.5 V	"	"	"			"	"	"	2_Y	"		"		"		"
		48	"	"			3.5 V	"	"	"	"			"	"	"	2_Y	"		"		"		"
		49	"	"			1.5 V	3.5 V	"	"	"			"	"	"	2_Y	"		"		"		"
		50	"	"			GND	GND	"	1.5 V	1.5 V			"	"	"	3_Y	"		"		"		"
		51	"	"			"	"	"	3.5 V	"			"	"	"	3_Y	"		"		"		"
		52	"	"			"	"	"	1.5 V	3.5 V			"	"	"	3_Y	"		"		"		"
		53	"	"			"	"	"	GND	GND			1.5 V	1.5 V	"	4_Y	"		"		"		"
		54	"	"			"	"	"	"	"			3.5 V	"	"	4_Y	"		"		"		"
		55	"	"			"	"	"	"	"			1.5 V	3.5 V	"	4_Y	"		"		"		"
V_{IL2}		56	3.0 V	3.0 V			"	"	"	"	"			GND	GND	10 V	1_Y	9.0		9.0		9.0		"
		57	7.0 V	"			"	"	"	"	"			"	"	"	1_Y	"		"		"		"
		58	3.0 V	7.0 V			"	"	"	"	"			"	"	"	1_Y	"		"		"		"
		59	GND	GND			3.0 V	3.0 V	"	"	"			"	"	"	2_Y	"		"		"		"
		60	"	"			7.0 V	"	"	"	"			"	"	"	2_Y	"		"		"		"
		61	"	"			3.0 V	7.0 V	"	"	"			"	"	"	2_Y	"		"		"		"
		62	"	"			GND	GND	"	3.0 V	3.0 V			"	"	"	3_Y	"		"		"		"
		63	"	"			"	"	"	7.0 V	"			"	"	"	3_Y	"		"		"		"
		64	"	"			"	"	"	3.0 V	7.0 V			"	"	"	3_Y	"		"		"		"
		65	"	"			"	"	"	GND	GND			3.0 V	3.0 V	"	4_Y	"		"		"		"
		66	"	"			"	"	"	"	"			7.0 V	"	"	4_Y	"		"		"		"
		67	"	"			"	"	"	"	"			3.0 V	7.0 V	"	4_Y	"		"		"		"
V_{IL3}		68	4.0 V	4.0 V			"	"	"	"	"			GND	GND	15 V	1_Y	13.5		13.5		13.5		"
		69	11 V	"			"	"	"	"	"			"	"	"	1_Y	"		"		"		"
		70	4.0 V	11 V			"	"	"	"	"			"	"	"	1_Y	"		"		"		"
		71	GND	GND			4.0 V	4.0 V	"	"	"			"	"	"	2_Y	"		"		"		"
		72	"	"			11 V	"	"	"	"			"	"	"	2_Y	"		"		"		"
		73	"	"			4.0 V	11 V	"	"	"			"	"	"	2_Y	"		"		"		"
		74	"	"			GND	GND	"	4.0 V	4.0 V			"	"	"	3_Y	"		"		"		"
		75	"	"			"	"	"	11 V	"			"	"	"	3_Y	"		"		"		"
		76	"	"			"	"	"	4.0 V	11 V			"	"	"	3_Y	"		"		"		"
		77	"	"			"	"	"	GND	GND			4.0 V	4.0 V	"	4_Y	"		"		"		"
		78	"	"			"	"	"	"	"			11 V	"	"	4_Y	"		"		"		"
		79	"	"			"	"	"	"	"			4.0 V	11 V	"	4_Y	"		"		"		"
I_{OL1}		80	5 V	5 V	0.4 V		"	"	"	"	"			GND	GND	5 V	1_Y	0.51		0.36		0.64		mA
		81	GND	GND		0.4 V	5 V	5 V	"	"	"			"	"	"	2_Y	"		"		"		"
		82	"	"			GND	GND	"	5 V	5 V	0.4 V		"	"	"	3_Y	"		"		"		"
		83	"	"			"	"	"	GND	GND		0.4 V	5 V	5 V	"	4_Y	"		"		"		"
I_{OL2}		84	15 V	15 V	1.5 V		"	"	"	"	"			GND	GND	15 V	1_Y	3.4		2.4		4.2		"
		85	GND	GND		1.5 V	15 V	15 V	"	"	"			"	"	"	2_Y	"		"		"		"
		86	"	"			GND	GND	"	15 V	15 V	1.5 V		"	"	"	3_Y	"		"		"		"
		87	"	"			"	"	"	GND	GND		1.5 V	15 V	15 V	"	4_Y	"		"		"		"
I_{OH1}		88	5.0 V	"	4.6 V		"	"	"	"	"			GND	GND	.5 V	1_Y	-0.51		-0.36		-0.64		"
		89	GND	5.0 V	4.6 V		"	"	"	"	"			"	"	"	1_Y	"		"		"		"
		90	"	GND		4.6 V	5.0 V	"	"	"	"			"	"	"	2_Y	"		"		"		"
		91	"	"		4.6 V	GND	5.0 V	"	"	"			"	"	"	2_Y	"		"		"		"
		92	"	"			"	GND	"	5.0 V	"	4.6 V		"	"	"	3_Y	"		"		"		"
		93	"	"			"	"	"	GND	5.0 V	4.6 V		"	"	"	3_Y	"		"		"		"
		94	"	"			"	"	"	"	GND		4.6 V	5.0 V	"	"	4_Y	"		"		"		"
		95	"	"			"	"	"	"	"		4.6 V	GND	5.0 V	"	4_Y	"		"		"		"

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, K, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1_A	1_B	1_Y	2_Y	2_B	2_A	V_{SS}	3_A	3_B	3_Y	4_Y	4_B	4_A	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{OH2}		96	15 V	GND	13.5 V		GND	GND	GND	GND	GND			GND	GND	15 V	1_Y	-3.4		-2.4		-4.2		mA
		97	GND	15 V	13.5 V		15 V	15 V									1_Y							
		98	"	GND		13.5 V	GND			15 V	GND	13.5 V		15 V			2_Y							
		99	"	"		13.5 V	"	"		GND	15 V	13.5 V		GND			2_Y							
		100	"	"			"			"	GND		13.5				3_Y							
		101	"	"			"						13.5				3_Y							
		102	"	"			"								15 V		4_Y							
		103	"	"			"								GND		4_Y							
I_{IH1} 2/	3010	104	18 V	18 V			18 V	18 V	"	18 V	18 V			18 V	18 V	18 V	All inputs together		8.0					mA
I_{IH2}	"	105	"	"			"	"	"	"	"			"	"	"	1_A		1.0		45			"
		106	"	"			"	"	"	"	"			"	"	"	1_B		"		"			
		107	"	"			"	"	"	"	"			"	"	"	2_B		"		"			
		108	"	"			"	"	"	"	"			"	"	"	2_A		"		"			
		109	"	"			"	"	"	"	"			"	"	"	3_A		"		"			
		110	"	"			"	"	"	"	"			"	"	"	3_B		"		"			
		111	"	"			"	"	"	"	"			"	"	"	4_B		"		"			
		112	"	"			"	"	"	"	"			"	"	"	4_A		"		"			
I_{IL1} 2/	3009	113	GND	GND			GND	GND	"	GND	GND			GND	GND	"	All inputs together		-8.0					"
I_{IL2}	"	114	"	18 V			18 V	18 V	"	18 V	18 V			18 V	18 V	"	1_A		-1.0		-45			"
		115	18 V	"			"	"	"	"	"			"	"	"	1_B		"		"			
		116	"	GND			GND	GND	"	"	"			"	"	"	2_B		"		"			
		117	"	18 V			18 V	18 V	"	GND	GND			"	"	"	2_A		"		"			
		118	"	"			"	"	"	18 V	18 V			"	"	"	3_A		"		"			
		119	"	"			"	"	"	"	"			"	"	"	3_B		"		"			
		120	"	"			"	"	"	"	"			GND	"	"	4_B		"		"			
		121	"	"			"	"	"	"	"			18 V	GND	"	4_A		"		"			
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_I	3012	122	3/	3/			3/	3/	GND	3/	3/			3/	3/	GND	1_A		12.0					pF
		123							"							"	1_B		"					
		124							"							"	2_B		"					
		125							"							"	2_A		"					
		126							"							"	3_A		"					
		127							"							"	3_B		"					
		128							"							"	4_B		"					
		129							"							"	4_A		"					

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	1_A	1_B	1_Y	2_Y	2_B	2_A	V_{SS}	3_A	3_B	3_Y	4_Y	4_B	4_A	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5	130	IN	5.0 V	OUT		5.0 V	5.0 V	GND	5.0 V	5.0 V			5.0 V	5.0 V	5.0 V	1_A to 1_Y	10	200	15	300	10	200	ns
		131	5.0 V	IN	OUT		"	"	"	"	"			"	"	"	1_B to 1_Y	"	"	"	"	"	"	"
		132	"	5.0 V		OUT	IN	"	"	"	"			"	"	"	2_B to 2_Y	"	"	"	"	"	"	"
		133	"	"		OUT	5.0 V	IN	"	"	"			"	"	"	2_A to 2_Y	"	"	"	"	"	"	"
		134	"	"			"	5.0 V	"	IN	"	OUT		"	"	"	3_A to 3_Y	"	"	"	"	"	"	"
		135	"	"			"	"	"	5.0 V	IN	OUT		"	"	"	3_B to 3_Y	"	"	"	"	"	"	"
		136	"	"			"	"	"	"	5.0 V		OUT	IN	"	"	4_B to 4_Y	"	"	"	"	"	"	"
		137	"	"			"	"	"	"	"		OUT	5.0 V	IN	"	4_A to 4_Y	"	"	"	"	"	"	"
t_{PLH}	"	138	IN	"	OUT		"	"	"	"	"			"	5.0 V	"	1_A to 1_Y	8	150	11	225	8	150	"
		139	5.0 V	IN	OUT		"	"	"	"	"			"	"	"	1_B to 1_Y	"	"	"	"	"	"	"
		140	"	5.0 V		OUT	IN	"	"	"	"			"	"	"	2_B to 2_Y	"	"	"	"	"	"	"
		141	"	"		OUT	5.0 V	IN	"	"	"			"	"	"	2_A to 2_Y	"	"	"	"	"	"	"
		142	"	"			"	5.0 V	"	IN	"	OUT		"	"	"	3_A to 3_Y	"	"	"	"	"	"	"
		143	"	"			"	"	"	5.0 V	IN	OUT		"	"	"	3_B to 3_Y	"	"	"	"	"	"	"
		144	"	"			"	"	"	"	5.0 V		OUT	IN	"	"	4_B to 4_Y	"	"	"	"	"	"	"
		145	"	"			"	"	"	"	"		OUT	5.0 V	IN	"	4_A to 4_Y	"	"	"	"	"	"	"
t_{THL}	3004 Fig. 5	146	IN	"	OUT		"	"	"	"	"			"	5.0 V	"	1_Y	15	300	23	450	15	300	"
		147	5.0 V	IN	OUT		"	"	"	"	"			"	"	"	1_Y	"	"	"	"	"	"	"
		148	"	5.0 V		OUT	IN	"	"	"	"			"	"	"	2_Y	"	"	"	"	"	"	"
		149	"	"		OUT	5.0 V	IN	"	"	"			"	"	"	2_Y	"	"	"	"	"	"	"
		150	"	"			"	5.0 V	"	IN	"	OUT		"	"	"	3_Y	"	"	"	"	"	"	"
		151	"	"			"	"	"	5.0 V	IN	OUT		"	"	"	3_Y	"	"	"	"	"	"	"
		152	"	"			"	"	"	"	5.0 V		OUT	IN	"	"	4_Y	"	"	"	"	"	"	"
		153	"	"			"	"	"	"	"		OUT	5.0 V	IN	"	4_Y	"	"	"	"	"	"	"
t_{TLH}	"	154	IN	"	OUT		"	"	"	"	"			"	5.0 V	"	1_Y	"	"	"	"	"	"	"
		155	5.0 V	IN	OUT		"	"	"	"	"			"	"	"	1_Y	"	"	"	"	"	"	"
		156	"	5.0 V		OUT	IN	"	"	"	"			"	"	"	2_Y	"	"	"	"	"	"	"
		157	"	"		OUT	5.0 V	IN	"	"	"			"	"	"	2_Y	"	"	"	"	"	"	"
		158	"	"			"	5.0 V	"	IN	"	OUT		"	"	"	3_Y	"	"	"	"	"	"	"
		159	"	"			"	"	"	5.0 V	IN	OUT		"	"	"	3_Y	"	"	"	"	"	"	"
		160	"	"			"	"	"	"	5.0 V		OUT	IN	"	"	4_Y	"	"	"	"	"	"	"
		161	"	"			"	"	"	"	"		OUT	5.0 V	IN	"	4_Y	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 52.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1_Y	1_A	1_B	1_C	1_D	NC	V_{SS}	NC	2_A	2_B	2_C	2_D	2_Y	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$ 1/		1 2 3 4 5 6 7 8		1 mA	1 mA	1 mA	1 mA				1 mA	1 mA	1 mA	1 mA		GND " " " " " " "	1_A 1_B 1_C 1_D 2_A 2_B 2_C 2_D		1.5 " " " " " " "					V " " " " " " "
$V_{IC(neg)}$ 1/		9 10 11 12 13 14 15 16		-1 mA	-1 mA	-1 mA	-1 mA		GND " " " " " "		-1 mA	-1 mA	-1 mA	-1 mA			1_A 1_B 1_C 1_D 2_A 2_B 2_C 2_D		-6.0 " " " " " " "					" " " " " " " "
I_{SS} 1/		17 18 19 20 21		GND 18.0 V " " "	18.0 V GND 18.0 V " "	18.0 V " GND 18.0 V "	18.0 V " " GND 18.0 V		" " " " "		GND 18.0 V " " "	18.0 V GND 18.0 V " "	18.0 V " GND 18.0 V "	18.0 V " " GND 18.0 V		18.0 V " " " "	V_{SS} " "		-25.0 " "		-750.0 " "			nA " "
V_{OH4}	3006 " " " " "	22 23 24 25 26 27 28 29		GND 15 V " " GND " " "	15 V GND 15 V " GND " " "	15 V " GND 15 V GND " " "	15 V " " GND " " " "		" " " " " " " "		GND " " " " 15 V " "	GND " " 15 V GND 15 V " "	GND " " 15 V " GND 15 V "	GND " " 15 V " " GND "		15.0 V " " " " " " "	1_Y 1_Y 1_Y 1_Y 2_Y 2_Y 2_Y 2_Y	14.95 " " " " " " "		14.95 " " " " " " "		14.95 " " " " " " "		V " " " " " " "
V_{OL4}	3007 "	30 31		15 V GND	15 V GND	15 V GND	15 V GND		" "		GND 15 V	GND 15 V	GND 15 V	" 15 V		" "	1_Y 2_Y		0.05 "		0.05 "		0.05 "	" "
V_{IH1}		32 33		3.5 V GND	3.5 V GND	3.5 V GND	3.5 V GND		" "		GND 3.5 V	GND 3.5 V	GND 3.5 V	GND 3.5 V		5 V "	1_Y 2_Y		0.5 "		0.5 "		0.5 "	" "
V_{IH2}		34 35		7.0 V GND	7.0 V GND	7.0 V GND	7.0 V GND		" "		GND 7.0 V	GND 7.0 V	GND 7.0 V	GND 7.0 V		10 V "	1_Y 2_Y		1.0 "		1.0 "		1.0 "	" "
V_{IH3}		36 37		11 V GND	11 V GND	11 V GND	11 V GND		" "		GND 11 V	GND 11 V	GND 11 V	GND 11 V		15 V "	1_Y 2_Y		1.5 "		1.5 "		1.5 "	" "
V_{IL1}		38 39 40 41 42 43 44 45		1.5 V 3.5 V " " GND " " "	3.5 V 1.5 V 3.5 V " GND " " "	3.5 V " 1.5 V 3.5 V GND " " "	3.5 V " " 1.5 V GND " " "		" " " " " " " "		GND " " 1.5 V 3.5 V " " "	GND " " 3.5 V 1.5 V 3.5 V " "	GND " " 3.5 V " 1.5 V 3.5 V "	GND " " 3.5 V " " 1.5 V "		5 V " " " " " " "	1_Y 1_Y 1_Y 1_Y 2_Y 2_Y 2_Y 2_Y	4.5 " " " " " " "		4.5 " " " " " " "		4.5 " " " " " " "		" " " " " " " "

See footnotes at end of device type 53.

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1_Y	1_A	1_B	1_C	1_D	NC	V_{SS}	NC	2_A	2_B	2_C	2_D	2_Y	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IL2}		46		3.0 V	7.0 V	7.0 V	7.0 V		GND		GND	GND	GND	GND		10 V	1_Y	9.0		9.0		9.0		V
		47		7.0 V	3.0 V	"	"		"		"	"	"	"		"	1_Y	"		"		"		"
		48		"	7.0 V	3.0 V	"		"		"	"	"	"		"	1_Y	"		"		"		"
		49		"	"	7.0 V	3.0 V		"		"	"	"	"		"	1_Y	"		"		"		"
		50		GND	GND	GND	GND		"		3.0 V	7.0 V	7.0 V	7.0 V		"	2_Y	"		"		"		"
		51		"	"	"	"		"		7.0 V	3.0 V	"	"		"	2_Y	"		"		"		"
		52		"	"	"	"		"		"	7.0 V	3.0 V	"		"	2_Y	"		"		"		"
		53		"	"	"	"		"		"	"	7.0 V	3.0 V		"	2_Y	"		"		"		"
V_{IL3}		54		4.0 V	11 V	11 V	11 V		"		GND	GND	GND	GND		15 V	1_Y	13.5		13.5		13.5		"
		55		11 V	4.0 V	"	"		"		"	"	"	"		"	1_Y	"		"		"		"
		56		"	11 V	4.0 V	"		"		"	"	"	"		"	1_Y	"		"		"		"
		57		"	"	11 V	4.0 V		"		"	"	"	"		"	1_Y	"		"		"		"
		58		GND	GND	GND	GND		"		4.0 V	11 V	11 V	11 V		"	2_Y	"		"		"		"
		59		"	"	"	"		"		11 V	4.0 V	"	"		"	2_Y	"		"		"		"
		60		"	"	"	"		"		"	11 V	4.0 V	"		"	2_Y	"		"		"		"
		61		"	"	"	"		"		"	"	11 V	4.0 V		"	2_Y	"		"		"		"
I_{OL1}		62	0.4 V	5.0 V	5.0 V	5.0 V	5.0 V		"		GND	GND	GND	GND		5.0 V	1_Y	0.51		0.36		0.64		mA
		63		GND	GND	GND	GND				5.0 V	5.0 V	5.0 V	5.0 V	0.4 V	"	2_Y	"		"		"		"
I_{OL2}		64	1.5 V	15 V	15 V	15 V	15 V		"		GND	GND	GND	GND		15 V	1_Y	3.4		2.4		4.2		"
		65		GND	GND	GND	GND				15 V	15 V	15 V	15 V	1.5 V	"	2_Y	"		"		"		"
I_{OH1}		66	4.6 V	"	5.0 V	5.0 V	5.0 V		"		GND	GND	GND	GND		5.0 V	1_Y	-0.51		-0.36		-0.64		"
		67	"	5.0 V	GND	"	"		"		"	"	"	"		"	1_Y	"		"		"		"
		68	"	"	5.0 V	GND	GND		"		"	"	"	"		"	1_Y	"		"		"		"
		69	"	"	"	5.0 V	"		"		"	"	"	"		"	1_Y	"		"		"		"
		70		GND	GND	GND	GND		"		"	5.0 V	5.0 V	5.0 V	4.6 V	"	2_Y	"		"		"		"
		71		"	"	"	"		"		5.0 V	GND	"	"	"	"	2_Y	"		"		"		"
		72		"	"	"	"		"		"	5.0 V	GND	"	"	"	2_Y	"		"		"		"
		73		"	"	"	"		"		"	"	5.0 V	GND		"	2_Y	"		"		"		"
I_{OH2}		74	13.5 V	"	15 V	15 V	15 V		"		GND	GND	GND	"		15 V	1_Y	-3.4		-2.4		-4.2		"
		75	"	15 V	GND	"	"		"		"	"	"	"		"	1_Y	"		"		"		"
		76	"	"	15 V	GND	GND		"		"	"	"	"		"	1_Y	"		"		"		"
		77	"	"	"	15 V	"		"		"	"	"	"		"	1_Y	"		"		"		"
		78		GND	GND	GND	GND		"		15 V	15 V	15 V	15 V	13.5 V	"	2_Y	"		"		"		"
		79		"	"	"	"		"		GND	GND	"	"	"	"	2_Y	"		"		"		"
		80		"	"	"	"		"		15 V	15 V	GND	"	"	"	2_Y	"		"		"		"
		81		"	"	"	"		"		"	"	15 V	GND		"	2_Y	"		"		"		"
I_{IH1} 2/	3010	82		18 V	18 V	18 V	18 V		"		18 V	18 V	18 V	18 V		18 V	All inputs together		8.0					nA
I_{IH2}	"	83		"	"	"	"		"		"	"	"	"		"	1_A		1.0		45			"
	"	84		"	"	"	"		"		"	"	"	"		"	1_B		"		"			"
	"	85		"	"	"	"		"		"	"	"	"		"	1_C		"		"			"
	"	86		"	"	"	"		"		"	"	"	"		"	1_D		"		"			"
	"	87		"	"	"	"		"		"	"	"	"		"	2_A		"		"			"
	"	88		"	"	"	"		"		"	"	"	"		"	2_B		"		"			"
	"	89		"	"	"	"		"		"	"	"	"		"	2_C		"		"			"
	"	90		"	"	"	"		"		"	"	"	"		"	2_D		"		"			"

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1_Y	1_A	1_B	1_C	1_D	NC	V_{SS}	NC	2_A	2_B	2_C	2_D	2_Y	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{IL1} 2/	3009	91		GND	GND	GND	GND		GND		GND	GND	GND	GND		18 V	All inputs together		-8.0					nA
I_{IL2}	"	92		"	18 V	18 V	18 V		"		18 V	18 V	18 V	18 V		"	1_A		-1.0		-45			"
	"	93		18 V	GND	"	"		"		"	"	"	"		"	1_B		"		"			
	"	94		"	18 V	GND	GND		"		"	"	"	"		"	1_C		"		"			
	"	95		"	"	18 V	18 V		"		"	"	"	"		"	1_D		"		"			
	"	96		"	"	"	"		"		GND	GND	GND	GND		"	2_A		"		"			
	"	97		"	"	"	GND		"		18 V	18 V	18 V	"		"	2_B		"		"			
	"	98		"	"	"	18 V		"		"	"	"	"		"	2_C		"		"			
	"	99		"	"	"	"		"		"	"	"	GND		"	2_D		"		"			

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 4 T_A = 25°C						
		Test no.	1_Y	1_A	1_B	1_C	1_D	NC	V_{SS}	NC	2_A	2_B	2_C	2_D	2_Y	V_{DD}		Min	Max					
C_I	3012	100		3/	3/	3/	3/		GND		3/	3/	3/	3/		GND	1_A		12.0					pF
	"	101							"							"	1_B		"					"
	"	102							"							"	1_C		"					"
	"	103							"							"	1_D		"					"
	"	104							"							"	2_A		"					"
	"	105							"							"	2_B		"					"
	"	106							"							"	2_C		"					"
	"	107							"							"	2_D		"					"

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	1_Y	1_A	1_B	1_C	1_D	NC	V_{SS}	NC	2_A	2_B	2_C	2_D	2_Y	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5	108	OUT	IN	5.0 V	5.0 V	5.0 V		GND		5.0 V	5.0 V	5.0 V	5.0 V		5.0 V	1_Y	16	325	25	490	15	325	ns
	"	109	"	5.0 V	IN	"	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
	"	110	"	"	5.0 V	IN	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
	"	111	"	"	"	5.0 V	IN		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
	"	112	"	"	"	"	5.0 V		"		"	"	"	"		"	2_Y	"	"	"	"	"	"	"
	"	113	"	"	"	"	"		"		"	"	"	"		"	2_Y	"	"	"	"	"	"	"
	"	114	"	"	"	"	"		"		"	"	"	"		"	2_Y	"	"	"	"	"	"	"
	"	115	"	"	"	"	"		"		"	"	"	"		"	2_Y	"	"	"	"	"	"	"
t_{PLH1}	"	116	OUT	IN	IN	IN	IN		"		IN	IN	IN	5.0 V	OUT	"	1_Y	13	250	19	375	13	250	"
	"	117	"	5.0 V	5.0 V	5.0 V	5.0 V		"		5.0 V	5.0 V	5.0 V	"	"	"	1_Y	"	"	"	"	"	"	"
	"	118	"	"	"	"	"		"		"	"	"	"	"	"	1_Y	"	"	"	"	"	"	"
	"	119	"	"	"	"	"		"		"	"	"	"	"	"	1_Y	"	"	"	"	"	"	"
	"	120	"	"	"	"	"		"		"	"	"	"	"	"	2_Y	"	"	"	"	"	"	"
	"	121	"	"	"	"	"		"		"	"	"	"	"	"	2_Y	"	"	"	"	"	"	"
	"	122	"	"	"	"	"		"		"	"	"	"	"	"	2_Y	"	"	"	"	"	"	"
	"	123	"	"	"	"	"		"		"	"	"	"	"	"	2_Y	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions and limits, see 1/ and 3/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	1_Y	1_A	1_B	1_C	1_D	NC	V_{SS}	NC	2_A	2_B	2_C	2_D	2_Y	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{THL}	3004 Fig. 5	124	OUT	IN	5.0 V	5.0 V	5.0 V		GND		5.0 V	5.0 V	5.0 V	5.0 V		5.0 V	1_Y	28	550	41	825	28	550	ns
		125	"	5.0 V	IN	"	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
		126	"	"	5.0 V	IN	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
		127	"	"	"	5.0 V	IN		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
		128		"	"	"	5.0 V		"		IN	"	"	"	OUT	"	2_Y	"	"	"	"	"	"	"
		129		"	"	"	"		"		5.0 V	IN	"	"	"	"	2_Y	"	"	"	"	"	"	"
		130		"	"	"	"		"		"	5.0 V	IN	"	"	"	2_Y	"	"	"	"	"	"	"
		131		"	"	"	"		"		"	"	5.0 V	IN	"	"	2_Y	"	"	"	"	"	"	"
t_{TLH}	"	132	OUT	IN	"	"	"		"		"	"	"	5.0 V		"	1_Y	21	425	32	640	21	425	"
	"	133	"	5.0 V	IN	"	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
	"	134	"	"	5.0 V	IN	"		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
	"	135	"	"	"	5.0 V	IN		"		"	"	"	"		"	1_Y	"	"	"	"	"	"	"
	"	136		"	"	"	5.0 V		"		IN	"	"	"	OUT	"	2_Y	"	"	"	"	"	"	"
	"	137		"	"	"	"		"		5.0 V	IN	"	"	"	"	2_Y	"	"	"	"	"	"	"
	"	138		"	"	"	"		"		"	5.0 V	IN	"	"	"	2_Y	"	"	"	"	"	"	"
	"	139		"	"	"	"		"		"	"	5.0 V	IN	"	"	2_Y	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 53.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	2_A	2_B	1_A	1_B	1_C	1_Y	V_{SS}	2_C	2_Y	3_Y	3_A	3_B	3_C	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$ 1/		1 2 3 4 5 6 7 8 9	1 mA	1 mA	1 mA	1 mA	1 mA			1 mA			1 mA	1 mA	1 mA	GND " " " " " " "	2_A 2_B 2_C 1_A 1_B 1_C 3_A 3_B 3_C		1.5 " " " " " " " "					V " " " " " " " "
$V_{IC(neg)}$ 1/		10 11 12 13 14 15 16 17 18	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA		GND " " " " " " "	-1 mA			-1 mA	-1 mA	-1 mA		2_A 2_B 2_C 1_A 1_B 1_C 3_A 3_B 3_C		-6.0 " " " " " " " "					" " " " " " " " "
I_{SS} 1/	3005 " "	19 20 21 22	GND 18 V "	18 V " GND 18 V	GND 18 V "	18 V " GND 18 V	18 V " " GND		" " " "	18 V " " GND			GND 18 V " "	18 V " GND 18 V	18 V " " GND	18 V " "	Y_{SS} " "		-25 " "		-750 " "			nA " "
V_{OH4}	3006 " " " " " " "	23 24 25 26 27 28 29 30 31	GND 15 V " GND " " " " "	15 V GND 15 V GND " " " " "	GND " " " 15 V GND " " "	GND " " 15 V GND 15 V GND " "	" " " 15 V " GND " " "		" " " " " " " " "	15 V " GND " " " " " "			GND " " " " " " 15 V "	GND " " " " " 15 V GND 15 V	" " " " " " 15 V " GND	15 V " " " " " " " "	2_Y 2_Y 2_Y 1_Y 1_Y 1_Y 3_Y 3_Y 3_Y	14.95 " " " " " " " "		14.95 " " " " " " " "		14.95 " " " " " " " "			V " " " " " " " "
V_{OL4}	3007 " "	32 33 34	15 V GND "	15 V GND "	" 15 V GND	" 15 V GND	" 15 V GND		" " "	15 V GND "			GND " 15 V	GND " 15 V	" " 15 V	" " "	2_Y 1_Y 3_Y		0.05 " "		0.05 " "		0.05 " "	" " "
V_{IH1}		35 36 37	3.5 V GND "	3.5 V GND "	" 3.5 V GND	" 3.5 V GND	" 3.5 V GND		" " "	3.5 V GND "			GND " 3.5 V	GND " 3.5 V	GND " 3.5 V	5 V " "	2_Y 1_Y 3_Y		0.5 " "		0.5 " "		0.5 " "	" " "
V_{IH2}		38 39 40	7.0 V GND "	7.0 V GND "	" 7.0 V GND	" 7.0 V GND	" 7.0 V GND		" " "	7.0 V GND "			GND " 7.0 V	GND " 7.0 V	GND " 7.0 V	10 V " "	2_Y 1_Y 3_Y		1.0 " "		1.0 " "		1.0 " "	" " "
V_{IH3}		41 42 43	11 V GND "	11 V GND "	" 11 V GND	" 11 V GND	" 11 V GND		" " "	11 V GND "			GND " 11 V	GND " 11 V	GND " 11 V	15 V " "	2_Y 1_Y 3_Y		1.5 " "		1.5 " "		1.5 " "	" " "

See footnotes at end of device type 53.

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions 1/														Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	2_A	2_B	1_A	1_B	1_C	1_Y	V_{SS}	2_C	2_Y	3_Y	3_A	3_B	3_C	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{TL1}		44	1.5 V	3.5 V	GND	GND	GND		GND	3.5 V			GND	GND	GND	5 V	2_Y	4.5		4.5		4.5		V
		45	3.5 V	1.5 V	"	"	"		"	"			"	"	"	"	2_Y	"		"		"		"
		46	"	3.5 V	"	"	"		"	1.5 V			"	"	"	"	2_Y	"		"		"		"
		47	GND	GND	1.5 V	3.5 V	3.5 V		"	GND			"	"	"	"	1_Y	"		"		"		"
		48	"	"	3.5 V	1.5 V	"		"	"			"	"	"	"	1_Y	"		"		"		"
		49	"	"	"	3.5 V	1.5 V		"	"			"	"	"	"	1_Y	"		"		"		"
		50	"	"	GND	GND	GND		"	"			1.5 V	3.5 V	3.5 V	"	3_Y	"		"		"		"
		51	"	"	"	"	"		"	"			3.5 V	1.5 V	"	"	3_Y	"		"		"		"
		52	"	"	"	"	"		"	"			"	3.5 V	1.5 V	"	3_Y	"		"		"		"
V_{TL2}		53	3.0 V	7.0 V	"	"	"		"	7.0 V			GND	GND	GND	10 V	2_Y	9.0		9.0		9.0		"
		54	7.0 V	3.0 V	"	"	"		"	"			"	"	"	"	2_Y	"		"		"		"
		55	"	7.0 V	"	"	"		"	3.0 V			"	"	"	"	2_Y	"		"		"		"
		56	GND	GND	3.0 V	7.0 V	7.0 V		"	GND			"	"	"	"	1_Y	"		"		"		"
		57	"	"	7.0 V	3.0 V	"		"	"			"	"	"	"	1_Y	"		"		"		"
		58	"	"	"	7.0 V	3.0 V		"	"			"	"	"	"	1_Y	"		"		"		"
		59	"	"	GND	GND	GND		"	"			3.0 V	7.0 V	7.0 V	"	3_Y	"		"		"		"
		60	"	"	"	"	"		"	"			7.0 V	3.0 V	"	"	3_Y	"		"		"		"
		61	"	"	"	"	"		"	"			"	7.0 V	3.0 V	"	3_Y	"		"		"		"
V_{TL3}		62	4.0 V	11 V	"	"	"		"	11 V			GND	GND	GND	15 V	2_Y	13.5		13.5		13.5		"
		63	11 V	4.0 V	"	"	"		"	"			"	"	"	"	2_Y	"		"		"		"
		64	"	11 V	"	"	"		"	4.0 V			"	"	"	"	2_Y	"		"		"		"
		65	GND	GND	4.0 V	11 V	11 V		"	GND			"	"	"	"	1_Y	"		"		"		"
		66	"	"	11 V	4.0 V	"		"	"			"	"	"	"	1_Y	"		"		"		"
		67	"	"	"	11 V	4.0 V		"	"			"	"	"	"	1_Y	"		"		"		"
		68	"	"	GND	GND	GND		"	"			4.0 V	11 V	11 V	"	3_Y	"		"		"		"
		69	"	"	"	"	"		"	"			11 V	4.0 V	"	"	3_Y	"		"		"		"
		70	"	"	"	"	"		"	"			"	11 V	4.0 V	"	3_Y	"		"		"		"
I_{OL1}		71	5.0 V	5.0 V	"	"	"		"	5.0 V	0.4 V		GND	GND	GND	5 V	2_Y	0.51		0.51		0.51		mA
		72	GND	GND	5.0 V	5.0 V	5.0 V	0.4 V	"	GND			"	"	"	"	1_Y	"		"		"		"
		73	"	"	GND	GND	GND		"	"		0.4 V	5.0 V	5.0 V	5.0 V	"	3_Y	"		"		"		"
I_{OL2}		74	15 V	15 V	"	"	"		"	15 V	1.5 V		GND	GND	GND	15 V	2_Y	3.4		2.4		4.2		"
		75	GND	GND	15 V	15 V	15 V	1.5 V	"	"			"	"	"	"	1_Y	"		"		"		"
		76	"	"	GND	GND	GND		"	"		1.5 V	15 V	15 V	15 V	"	3_Y	"		"		"		"
I_{OH1}		77	"	5.0 V	"	"	"		"	5.0 V	4.6 V		GND	GND	GND	5 V	2_Y	-0.51		-0.36		-0.64		"
		78	5.0 V	GND	"	"	"		"	"	"		"	"	"	"	2_Y	"		"		"		"
		79	"	5.0 V	"	"	"		"	GND			"	"	"	"	2_Y	"		"		"		"
		80	GND	GND	"	5.0 V	5.0 V	4.6 V	"	"			"	"	"	"	1_Y	"		"		"		"
		81	"	"	5.0 V	GND	"	"	"	"			"	"	"	"	1_Y	"		"		"		"
		82	"	"	"	5.0 V	GND		"	"			"	"	"	"	1_Y	"		"		"		"
		83	"	"	GND	GND	"		"	"		4.6 V	"	5.0 V	5.0 V	"	3_Y	"		"		"		"
		84	"	"	"	"	"		"	"		"	5.0 V	GND	"	"	3_Y	"		"		"		"
		85	"	"	"	"	"		"	"		"	"	5.0 V	GND	"	3_Y	"		"		"		"
I_{OH2}		86	"	15 V	"	"	"		"	15 V	13.5 V		GND	GND	"	15 V	2_Y	-3.4		-2.4		-4.2		"
		87	15 V	GND	"	"	"		"	"	"		"	"	"	"	2_Y	"		"		"		"
		88	"	15 V	"	"	"		"	GND			"	"	"	"	2_Y	"		"		"		"
		89	GND	GND	"	"	"	13.5 V	"	"			"	"	"	"	1_Y	"		"		"		"
		90	"	"	15 V	15 V	15 V	"	"	"			"	"	"	"	1_Y	"		"		"		"
		91	"	"	"	GND	GND		"	"			"	"	"	"	1_Y	"		"		"		"
		92	"	"	GND	15 V	"		"	"		13.5 V	"	15 V	15 V	"	3_Y	"		"		"		"
		93	"	"	"	GND	"		"	"		"	15 V	GND	"	"	3_Y	"		"		"		"
		94	"	"	"	"	"		"	"		"	"	15 V	GND	"	3_Y	"		"		"		"

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	2_A	2_B	1_A	1_B	1_C	1_Y	Y_{SS}	2_C	2_Y	3_Y	3_A	3_B	3_C	Y_{DD}		Min	Max	Min	Max	Min	Max	
I_{IH1} 2/	3010	95	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V		GND	18.0 V			18.0 V	18.0 V	18.0 V	18.0 V	All together		9.0					nA
I_{IH2}	"	96	"	"	"	"	"		"	"			"	"	"	"	2_A		1.0		45.0			"
	"	97	"	"	"	"	"		"	"			"	"	"	"	2_B		"		"			"
	"	98	"	"	"	"	"		"	"			"	"	"	"	2_C		"		"			"
	"	99	"	"	"	"	"		"	"			"	"	"	"	1_A		"		"			"
	"	100	"	"	"	"	"		"	"			"	"	"	"	1_B		"		"			"
	"	101	"	"	"	"	"		"	"			"	"	"	"	1_C		"		"			"
	"	102	"	"	"	"	"		"	"			"	"	"	"	3_A		"		"			"
	"	103	"	"	"	"	"		"	"			"	"	"	"	3_B		"		"			"
	"	104	"	"	"	"	"		"	"			"	"	"	"	3_C		"		"			"
I_{IL1} 2/	3009	105	GND	GND	GND	GND	GND		"	GND			GND	GND	GND	"	All together		-9.0					"
I_{IL2}	"	106	"	18.0 V	18.0 V	18.0 V	18.0 V		"	18.0 V			18.0 V	18.0 V	18.0 V	"	2_A		-1.0		-45.0			"
	"	107	18.0 V	GND	"	"	"		"	"			"	"	"	"	2_B		"		"			"
	"	108	"	18.0 V	GND	GND	GND		"	GND			"	"	"	"	2_C		"		"			"
	"	109	"	"	18.0 V	18.0 V	18.0 V		"	18.0			"	"	"	"	1_A		"		"			"
	"	110	"	"	"	"	"		"	"			"	"	"	"	1_B		"		"			"
	"	111	"	"	"	"	"		"	"			"	"	"	"	1_C		"		"			"
	"	112	"	"	"	"	"		"	"			GND	GND	18.0 V	"	3_A		"		"			"
	"	113	"	"	"	"	"		"	"			18.0 V	18.0 V	GND	"	3_B		"		"			"
	"	114	"	"	"	"	"		"	"			"	"	"	"	3_C		"		"			"
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_1	3012	115	3/		3/				GND							GND	2_A		12.0					pF
	"	116		3/					"	3/						"	2_B		"					"
	"	117							"							"	2_C		"					"
	"	118			3/				"							"	1_A		"					"
	"	119				3/			"							"	1_B		"					"
	"	120					3/		"							"	1_C		"					"
	"	121							"				3/			"	3_A		"					"
	"	122							"					3/		"	3_B		"					"
	"	123							"						3/	"	3_C		"					"
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5	124	IN	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	OUT	GND	5.0 V	OUT		5.0 V	5.0 V	5.0 V	5.0 V	2_Y	9	175	13	265	9	175	ns
	"	125	5.0 V	IN	5.0 V	"	"	"	"	"	IN	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
	"	126	"	5.0 V	IN	5.0 V	"	"	"	"	5.0 V	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
	"	127	"	"	5.0 V	IN	5.0 V	"	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
	"	128	"	"	"	5.0 V	IN	"	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
	"	129	"	"	"	"	5.0 V	"	"	"	"	"		"	"	"	"	1_Y	"	"	"	"	"	"	"
	"	130	"	"	"	"	"	"	"	"	"	"		"	"	"	"	3_Y	"	"	"	"	"	"	"
	"	131	"	"	"	"	"	"	"	"	"	"		"	"	"	"	3_Y	"	"	"	"	"	"	"
	"	132	"	"	"	"	"	"	"	"	"	"		"	"	"	"	3_Y	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	For terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	2_A	2_B	1_A	1_B	1_C	1_Y	V_{SS}	2_C	2_Y	3_Y	3_A	3_B	3_C	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PLH}	3003 Fig. 5	133	IN	5.0 V	5.0 V	5.0 V	5.0 V		GND	5.0 V	OUT		5.0 V	5.0 V	5.0 V	5.0 V	2_Y	8	150	11	225	8	150	ns
		134	5.0 V	IN	"	"	"		"	"	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		135	"	5.0 V	"	"	"		"	IN	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		136	"	"	IN	"	"	OUT	"	5.0 V			"	"	"	"	1_Y	"	"	"	"	"	"	"
		137	"	"	5.0 V	IN	"	"	"	"			"	"	"	"	1_Y	"	"	"	"	"	"	"
		138	"	"	"	5.0 V	IN	"	"	"			"	"	"	"	1_Y	"	"	"	"	"	"	"
		139	"	"	"	"	5.0 V		"	"		OUT	IN	"	"	"	3_Y	"	"	"	"	"	"	"
		140	"	"	"	"	"		"	"		"	5.0 V	IN	"	"	3_Y	"	"	"	"	"	"	"
		141	"	"	"	"	"		"	"		"	"	5.0 V	IN	"	3_Y	"	"	"	"	"	"	"
t_{THL}	3004 Fig. 5	142	IN	"	"	"	"		"	"	OUT		"	"	5.0 V	"	2_Y	13	250	19	375	13	250	"
		143	5.0 V	IN	"	"	"		"	"	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		144	"	5.0 V	"	"	"		"	IN	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		145	"	"	IN	"	"	OUT	"	5.0 V			"	"	"	"	1_Y	"	"	"	"	"	"	"
		146	"	"	5.0 V	IN	"	"	"	"			"	"	"	"	1_Y	"	"	"	"	"	"	"
		147	"	"	"	5.0 V	IN	"	"	"			"	"	"	"	1_Y	"	"	"	"	"	"	"
		148	"	"	"	"	5.0 V		"	"		OUT	IN	"	"	"	3_Y	"	"	"	"	"	"	"
		149	"	"	"	"	"		"	"		"	5.0 V	IN	"	"	3_Y	"	"	"	"	"	"	"
		150	"	"	"	"	"		"	"		"	"	5.0 V	IN	"	3_Y	"	"	"	"	"	"	"
t_{TLH}	"	151	IN	"	"	"	"		"	"	OUT		"	"	5.0 V	"	2_Y	15	300	25	450	12	300	"
		152	5.0 V	IN	"	"	"		"	"	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		153	"	5.0 V	"	"	"		"	IN	"		"	"	"	"	2_Y	"	"	"	"	"	"	"
		154	"	"	IN	"	"	OUT	"	5.0 V			"	"	"	"	1_Y	"	"	"	"	"	"	"
		155	"	"	5.0 V	IN	"	"	"	"			"	"	"	"	1_Y	"	"	"	"	"	"	"
		156	"	"	"	5.0 V	IN	"	"	"			"	"	"	"	1_Y	"	"	"	"	"	"	"
		157	"	"	"	"	5.0 V		"	"		OUT	IN	"	"	"	3_Y	"	"	"	"	"	"	"
		158	"	"	"	"	"		"	"		"	5.0 V	IN	"	"	3_Y	"	"	"	"	"	"	"
		159	"	"	"	"	"		"	"		"	"	5.0 V	IN	"	3_Y	"	"	"	"	"	"	"

1/ Pins not designated may be "High" level logic, "Low" level logic, or open. Exceptions are as follows:

a. $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open.
b. $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open.
c. I_{SS} tests, the output terminal shall be open.

2/ The device manufacturer may at his option measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.
3/ See 4.4.1c.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 4, or equivalent, shall be used.

b. A special subgroup shall be added using an LTPD of 15 for classes S and B, and shall be performed on each inspection lot for initially qualified device types 01, 02, 03, and measured only for initial qualification and after process or design changes for initially qualified device types 51, 52, 53. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

c. End-point electrical parameters shall be as specified in table II herein and shall consist of those subgroups specified in table IIa of test method 5005 of MIL-STD-883. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 and as shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits as specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal, unless otherwise specified. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When these tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to a temperature of 25°C ±3°C; then, electrical parameter end-point measurements shall be performed.

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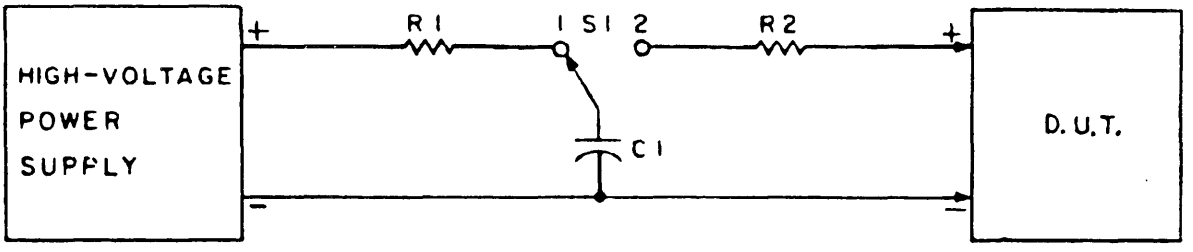
TABLE IV. Delta limits @25°C.

Parameter 1/	Device types	
	01, 02, 03	51, 52, 53
I_{SS}	±10 nA	±10 nA
V_{OL1}	±.04 V	---
V_{OH1}	±.08 V	---
I_{OL1}	---	±15%
I_{OH1}	---	±15%

1/ Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine delta's (Δ).

4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of 4) of the DUT shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit on figure 6.

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at the specified V_{DD}. Measure I_{SS} on the DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at the specified V_{DD}.



V_{ZAP} = 400 V minimum charge on C1
1 MΩ ≤ R1 ≤ 50 MΩ
R2 = 1.5 kΩ
C1 = 100 pF
S1 = Hg-wetted "bounceless" relay

FIGURE 6. High voltage (V_{ZAP}) test circuits.

b. V_{ZAP} is applied to DUT in the following modes (see table V) by changing C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated output

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c. Within 24 hours repeat the I_{SS}, I_{IL}, and I_{IH} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS} test). When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end-point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end-point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table VIII and figure 7. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardening end-point electrical parameters at 25°C.

Parameter	All device types	V_{DD}	
		Device types 01-03	Device types 51-53
V_{TN}	0.3 V min	10 V	10 V
V_{TP}	2.8 V max	10 V	10 V
ΔV_T	1.4 V max	10 V	10 V
I_{SS}	100 x max limit	15 V	18 V
t_{PLH}	1.35 x max limit	5 V	5 V
t_{PHL}	1.35 x max limit	5 V	5 V

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	V_{DD} = 10 V dc (through a 30- to 60 kilohm resistor)	V_{SS} = GND	V_{DD} = 10 V dc
01, 51	1, 2, 5, 6, 8, 9, 12, 13	7	14
02, 52	2, 3, 4, 5, 9, 10, 11, 12	7	14
03, 53	1, 2, 3, 4, 5, 8, 11, 12, 13	7	14

Pins not designated are open or connected to 10 V dc through a 30- to 60 kilohm resistor.

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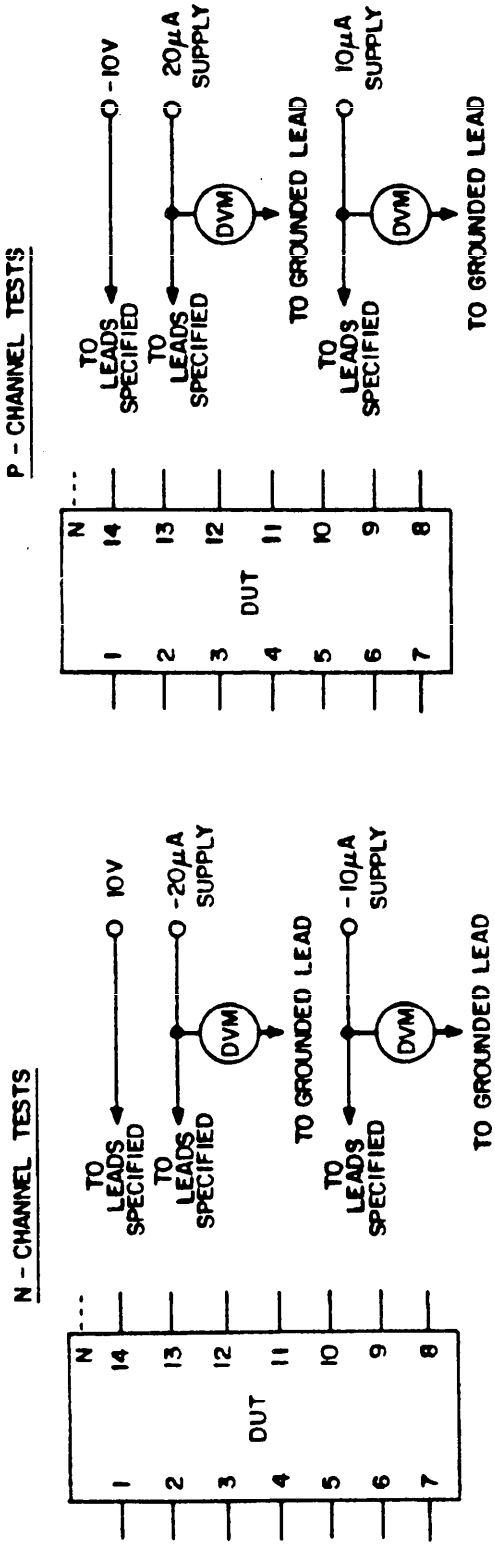


FIGURE 7. Threshold voltage test circuit.

TABLE VIII. Threshold - voltage test circuit conditions.

Device type	GND	10 V	V_{TN} measured at -20 μA supply	V_{TN} measured at -10 μA supply	GND	-10 V	V_{TP} measured at 20 μA supply	V_{TP} measured at 10 μA supply
01, 51	2	1, 14		5-9, 12, 13	2	5-9, 12 13		1, 14
02, 52	2	3-5, 14		7, 9-12	2	7, 9-12		3-5, 14
03, 53	3	4, 5, 14		1, 2, 7, 8, 11-13	3	1, 2, 7, 8, 11-13		4, 5, 14

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4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied.

a. Attributes data for all screening tests (see 4.2 and variables data for all static burn-in, dynamic burn-in, and steady state life tests (see 3.5).

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parmeters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

i. Requirements for total dose radiation testing (see 3.6.1 and 4.5.5), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

C_I	Input terminal-to-V_{SS} capacitance.
GND	Ground. Zero voltage potential.
I_{SS}	Quiescent supply current.
T_A	Free air temperature.
V_{DD}	Positive supply voltage.
V_{SS}	Negative supply voltage.
V_{ZAP}	Input test voltage.

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6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	4011A
02	4012A
03	4023A
51	4011B
52	4012B
53	4023B

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static build up. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM, AR
Navy - AS, CG, OC, MC, SH

Agent:
DLA - ES

Preparing activity:
NASA - NA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/50E	MICROCIRCUITS, DIGITAL, CMOS, NAND GATES MONOLITHIC SILICON, POSITIVE LOGIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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